The Solve it screen as built (2026-10-13)

What this is. A record of what the Solve it screens actually do today, for the design chat to work from. Nothing in the app was changed. Every statement is **observed** (a screenshot, a measurement, or a file and line) or marked **inferred** with the reason.

How it was captured. Commit `7d1485f` (main, 2026-10-13). The app ran on the local dev server; pages were driven by Playwright in the pre-installed Chromium, in a **1366×768 window** (key screens also at 1280×720). Fixed seeds (`?pack=add-both&level=2&seed=7&nosignin=1`) so every screen can be reproduced. Measurements are in `measure.json`, colours of the pieces in `pieces.json`, the list of every shot in `manifest.json`.

> **Blind spot to keep in mind.** A Chromebook *screen* is 1366×768, but the page only gets what is left after Chrome's tab strip and address bar (about **1366×650**). Every "fits / doesn't fit" below is for a 768-px-tall page, so the real margin on a student's Chromebook is roughly 120 px worse. Worth one test on a real school Chromebook before any layout decision.

Headlines (what surprised me)

1. **The two cards are built on different engines.** Divide it runs on the shared Solve it loop (`src/engine/solveLoop.js`). **Add to both sides does not:** it has its own reducer (`src/engine/addBothMoves.js`) from before the loop existed. So on Add the choice is still fixed per problem, there is no folding and no Next after a move, Start over follows the old three-red rule, and the scaffold replaces the work stack. Any change to "the loop" today means changing two code paths, or moving Add onto the loop first.
2. **Add to both sides does not fit 1366×768.** The page is 955 px tall; Undo is cut in half at the bottom edge and the help row and number pad are only reachable by scrolling (`add-02`). With I'm stuck open it is 1,100 px. Divide it fits exactly (768 px).
3. **On Add, six of the eight Why? lights turn green together with Fewer terms** although nothing judges them (`add-11`). Only Fewer terms and Solved for x have rules; the others copy Fewer terms through a fallback in `src/view/lightScreen.js:287`. The Add spec (§3E) says only those two boxes should show.
4. **There is no "partly" (yellow) light anywhere.** The only in-between state is Add's grey "–" (Fewer terms after a partly cancelling move, `add-20`), which means "no change", not "progress".
5. **Divide it's lights are not computed from the equation.** Any successful fraction sets Smaller numbers green as a constant (`src/engine/divideMoves.js:227`); One box is "coefficient is 1". Several lights (No fractions, No opposite box, x on the left, Each side simplified) have no rule anywhere in the code.

6. **Nothing previews a move today.** Hovering a live move shows only its name in a tooltip (`add-07`, `div-04`); nothing appears on the equation. On a touchscreen one tap selects the move at once and shows no name (`touch-01`), which is also short of loop §5 ("a tap shows its text for a moment before it acts").
7. **A pilot of idea 7 already exists:** `?layout=tucked` on Divide it (the counters tray slides up only when asked for; the moves sit in a drawer; the lights stand alone in a slim column). It has two bugs I could see: after a move, **the keyboard stops typing** into `x =` (focus lands on the Moves tab, and `src/main.js:541` ignores keys while a non-pad button has focus), and the counters **tray is clipped** at 1366×768 (`divt-24` to `divt-27`). **Update 2026-10-15: both are fixed** (focus now stays on the page after a move, so typing and Enter reach `x =` and Check; the tray lays its drawing and the fractions side by side and fits 1366×768 and 1280×720). The screenshots in this file are from before the fix.
8. **Start over on Add never unlocks after three red moves if the student presses Undo each time** (the spec's own flow): Undo restores the red count saved before the move (`addBothMoves.js:131`), so it never reaches three (`add-23`, `add-24`).
9. **Name clash:** Count it already has a card called **Flip It** (pack id `flipit`, `src/packs/index.js:4`). The Solve it card spec'd as "Flip it" will collide in name and, if reused, in id and save bits.
10. **The bold text is not as bold as the CSS asks.** Only Lexend 400 and 600 are loaded (`index.html:16`), while the line and buttons ask for 700-800. *Inferred:* the browser draws them with the 600 face. If "large and bold" matters, load 700/800.

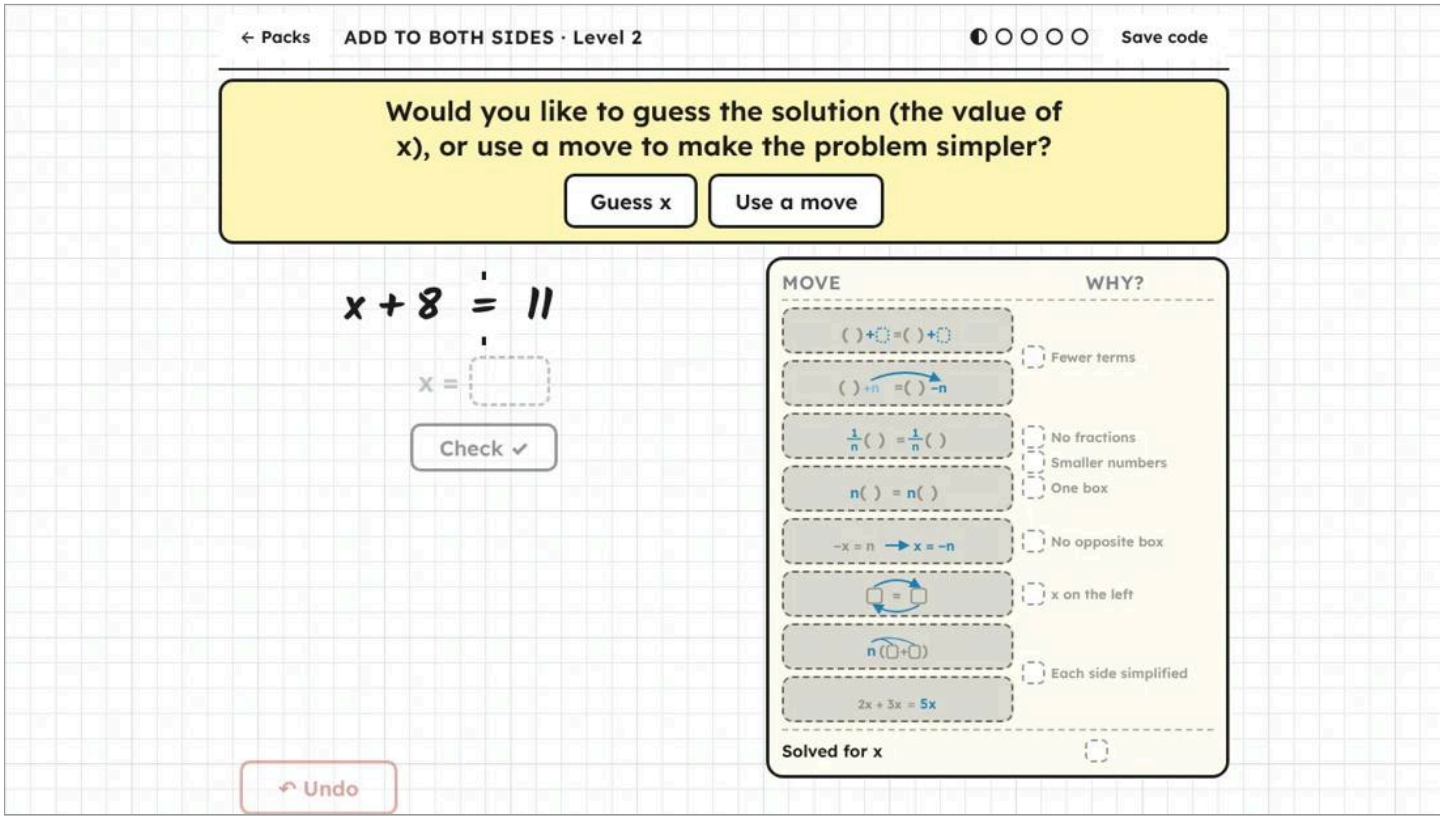
1.1 Screenshots

All shots are in `shots/` (WebP), listed by state below and embedded in `CURRENT-STATE.html`. Prefixes: `add-` Add to both sides, `div-` Divide it (classic, the default), `divt-` Divide it with `?layout=tucked`, `touch-` touchscreen emulation. A shot marked *scrolled* has a `-top` twin: the same moment before scrolling.

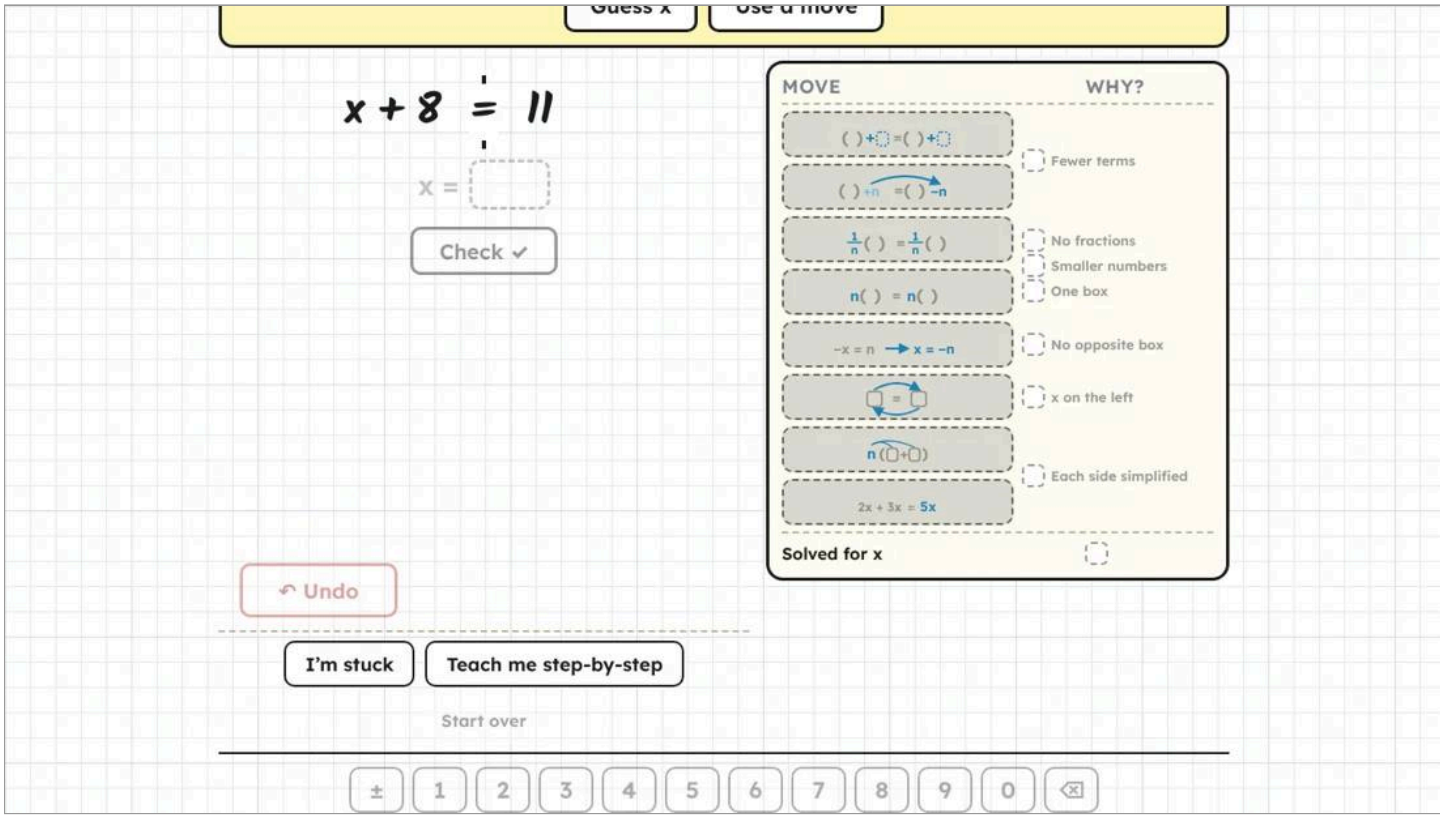
Not captured, and why: Flip it is not built (spec only). "Pressing" a move looks the same as hovering (there is no pressed style). Divide it's L1 final walkthrough was not reached in the script (the five items were). An opposite box is not drawn by any Solve it card (see 1.4).

This PDF shows 25 key screens. All 164 screenshots, with every state, are in the full `CURRENT-STATE.html` in the repo (`docs/lights-and-moves/`).

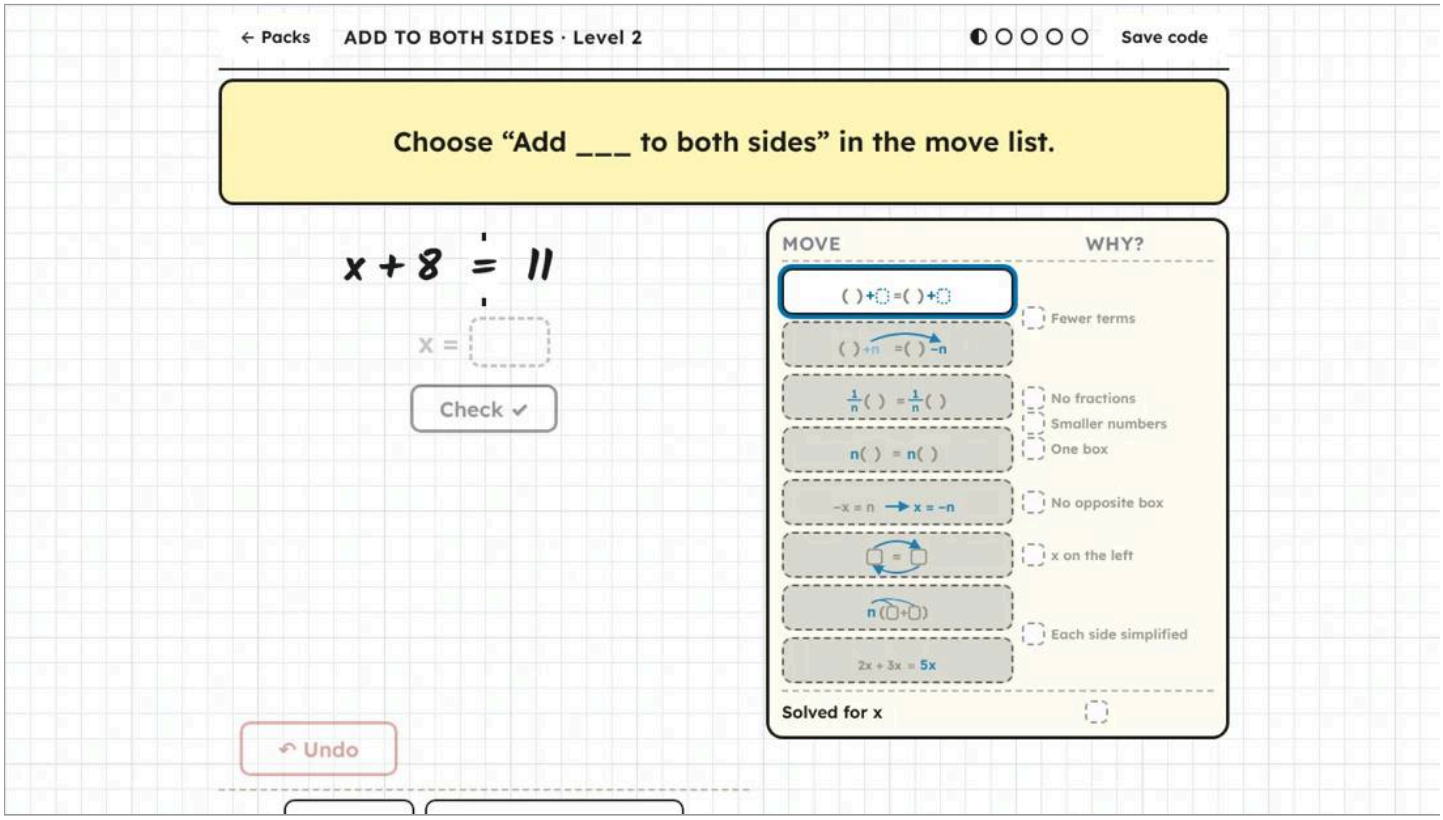
add-01 Add L2 · opening: the choice; moves locked



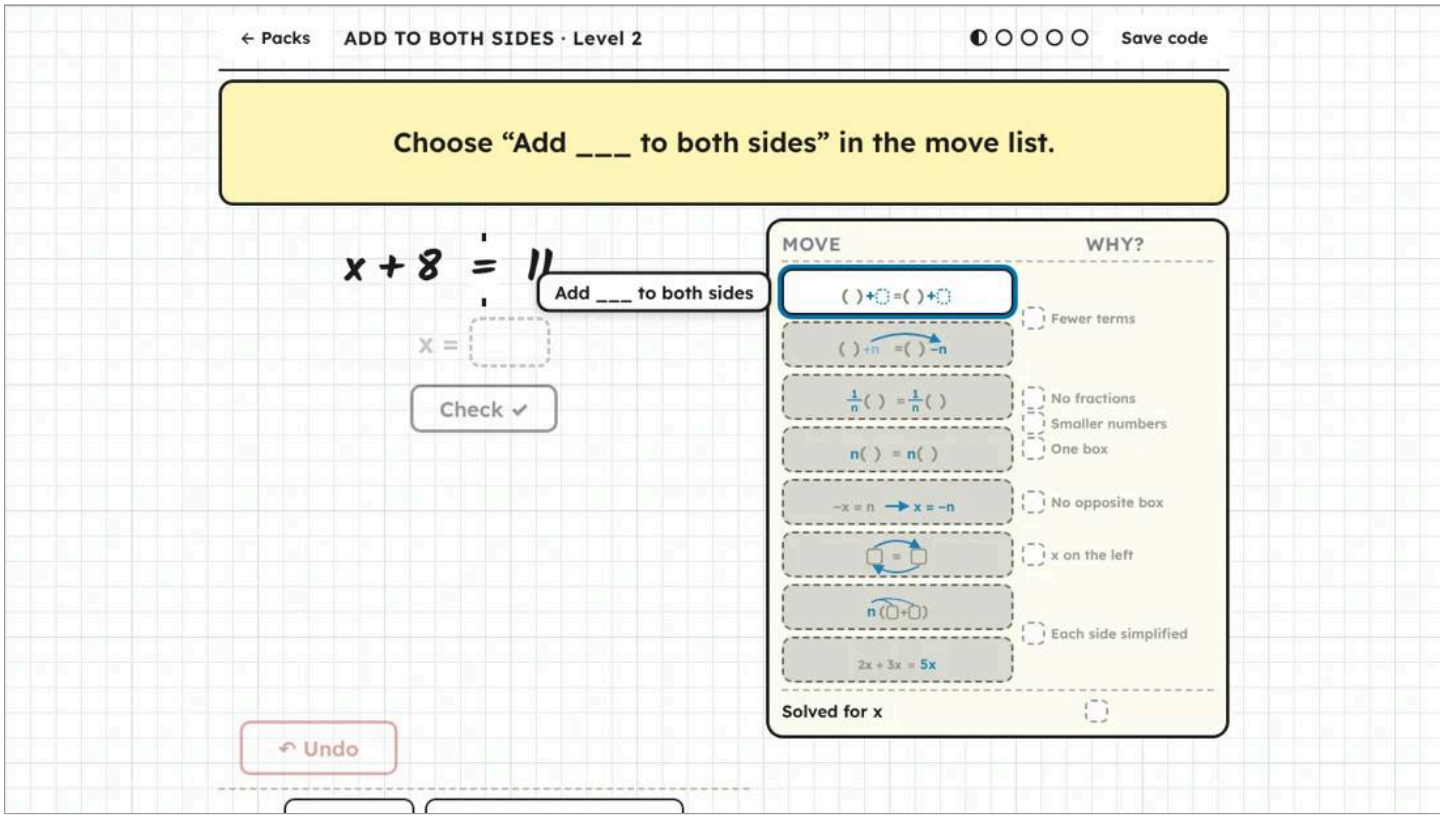
add-02 Add L2 · scrolled: help row and pad are below the 768-px edge



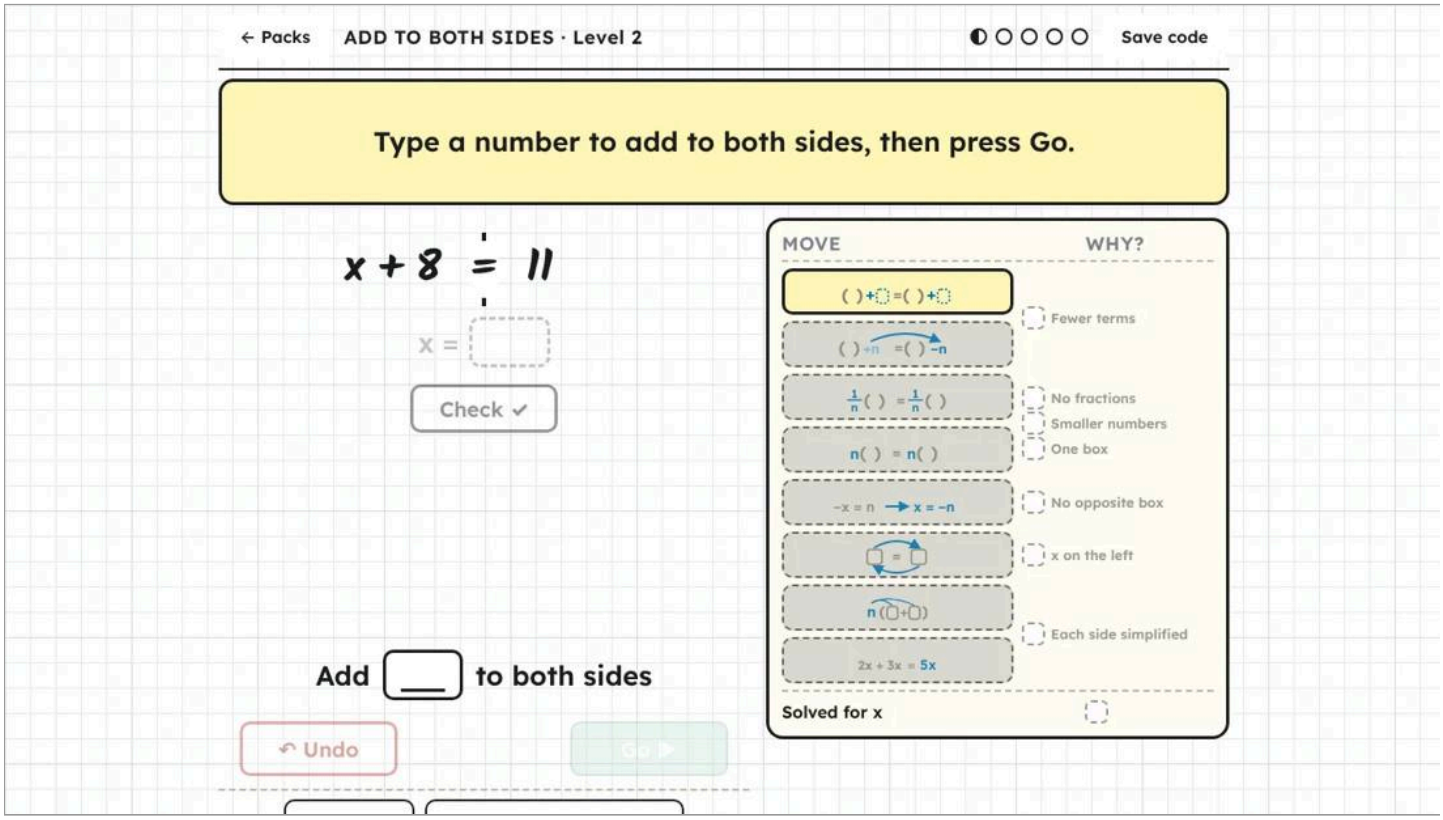
add-06 Add L2 · after Use a move



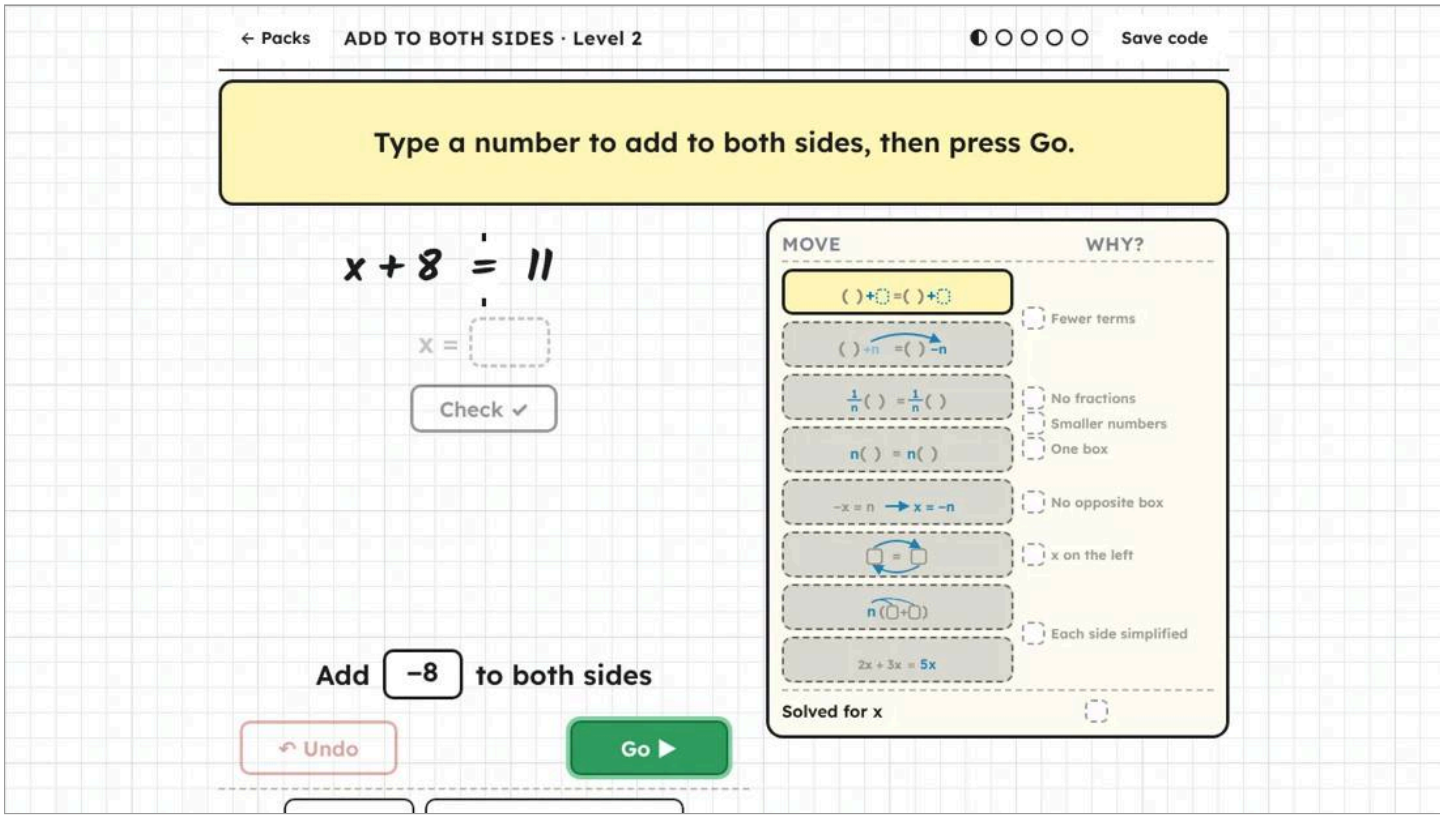
add-07 Add L2 · hovering Add: only a name tooltip



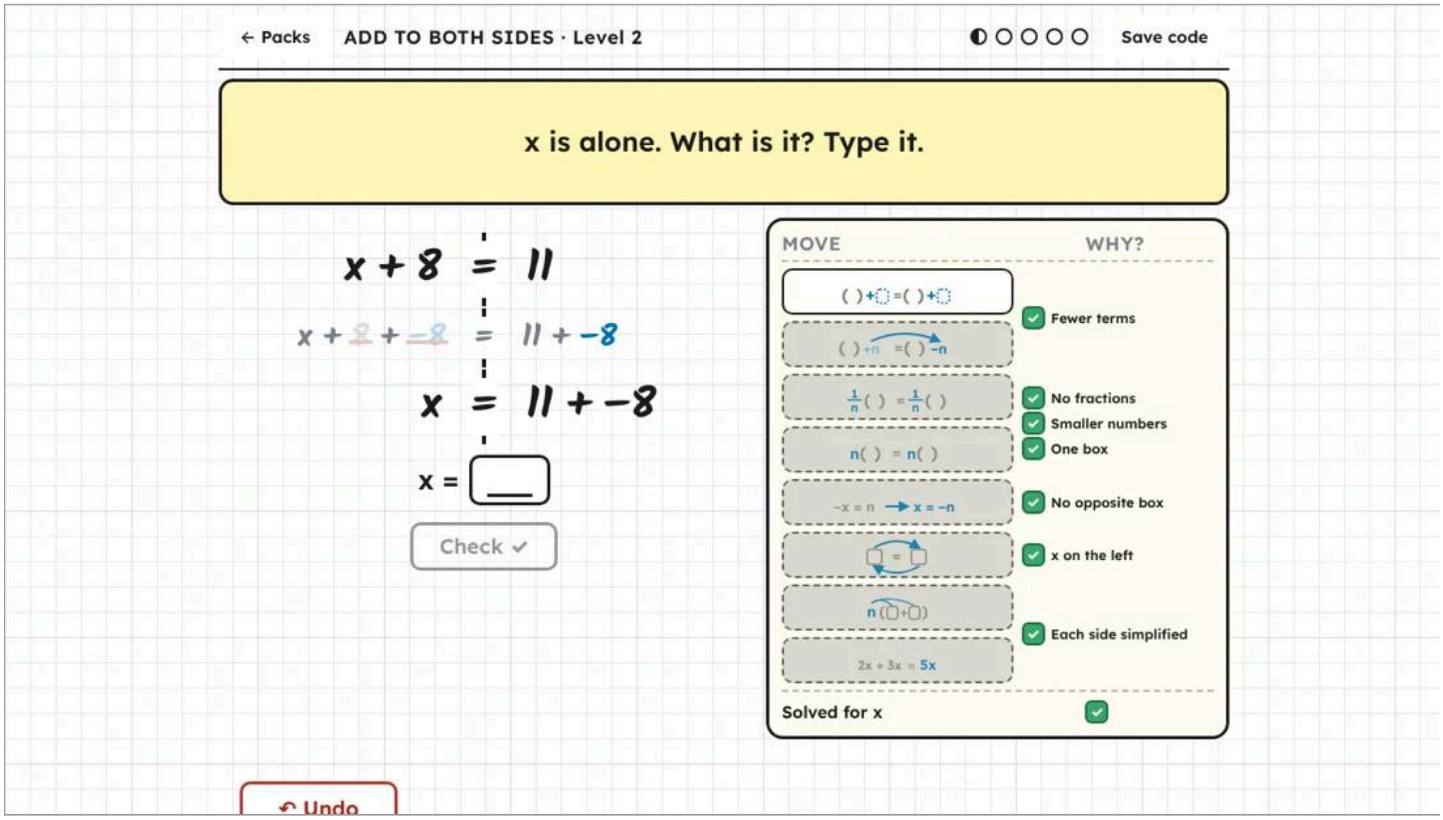
add-09 Add L2 · Add chosen: the reserved “Add [ ] to both sides” row



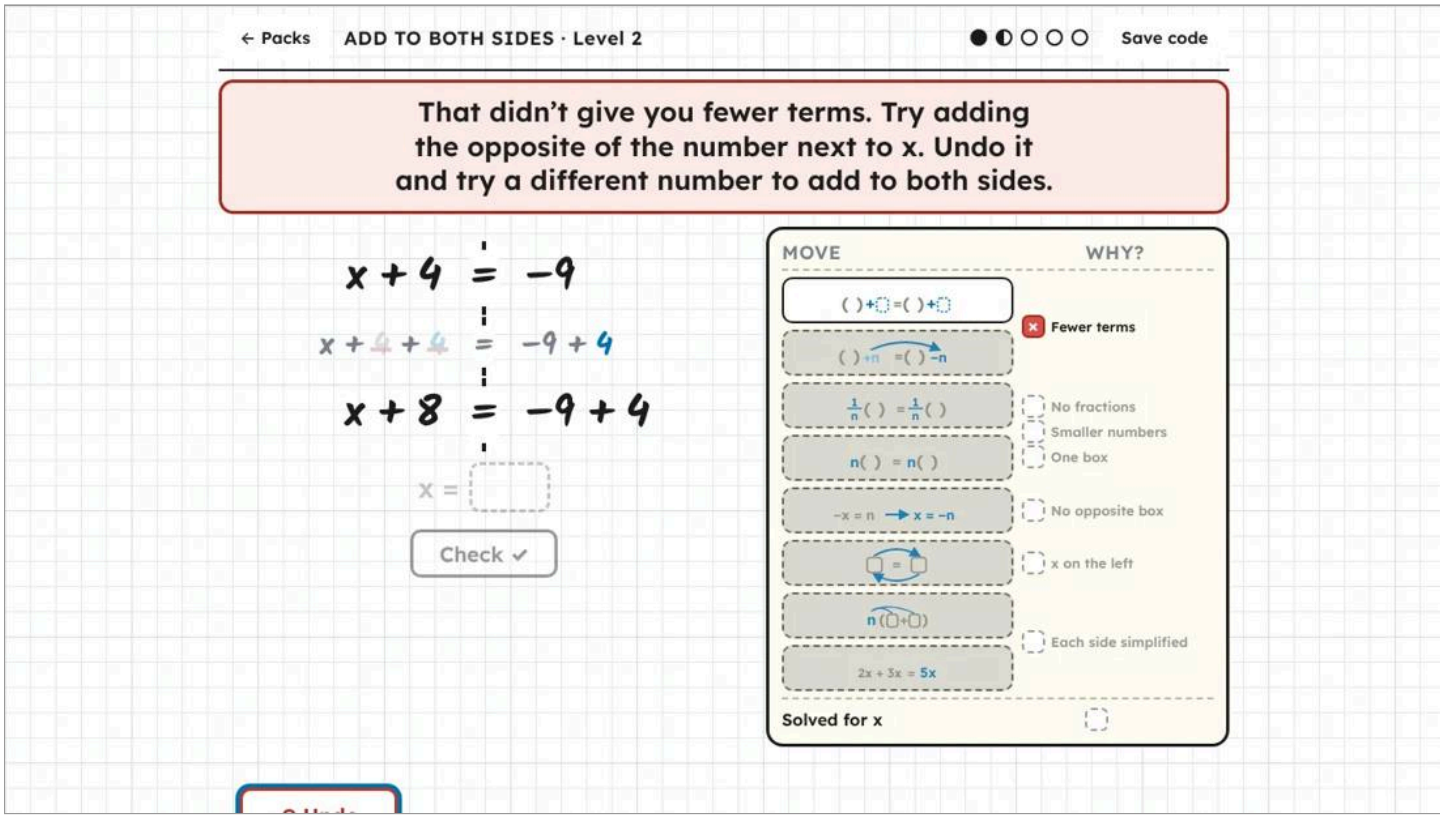
add-10 Add L2 · amount typed, Go pulses



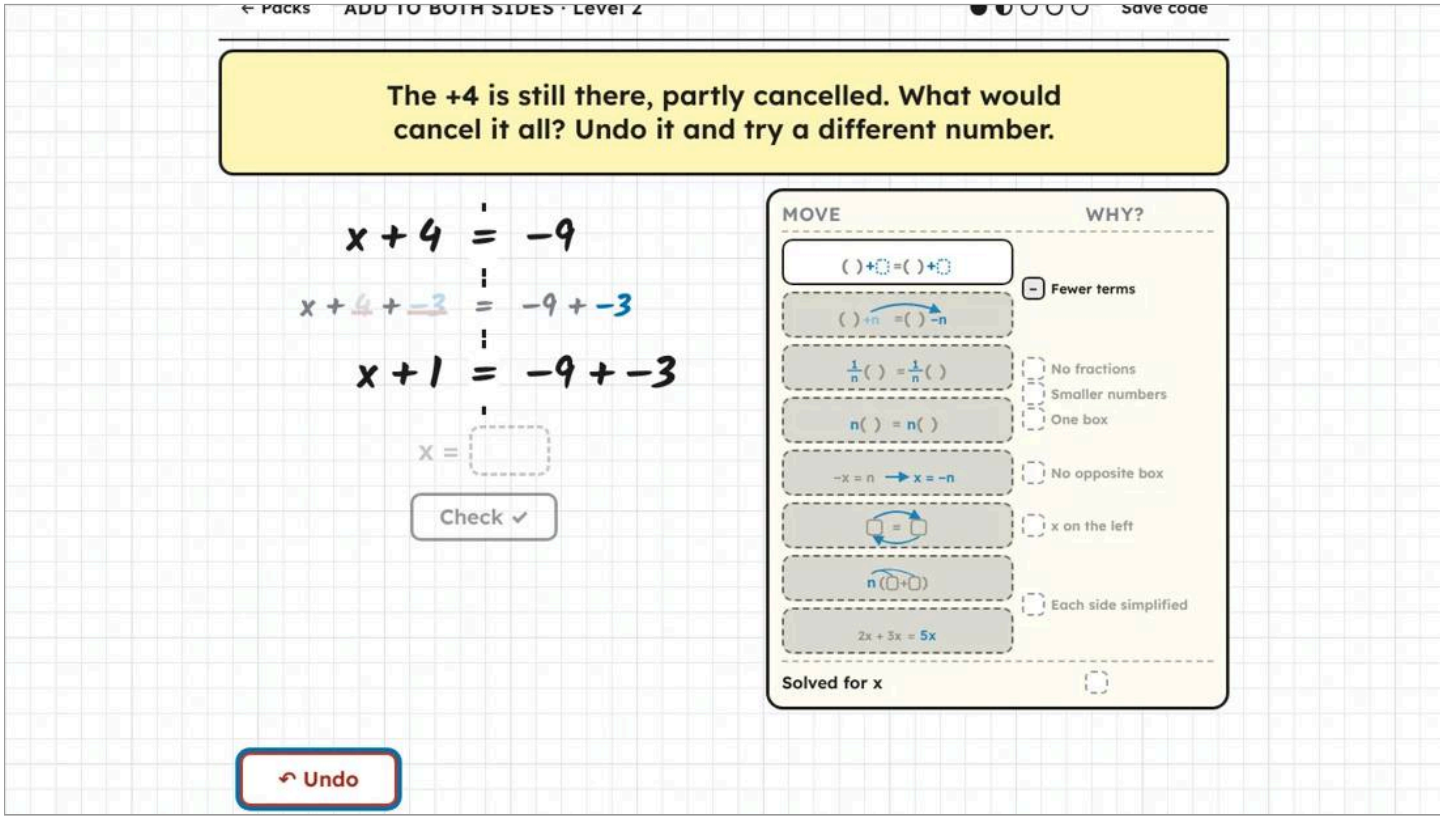
add-11 Add L2 · helpful move: six unjudged lights turn green with Fewer terms



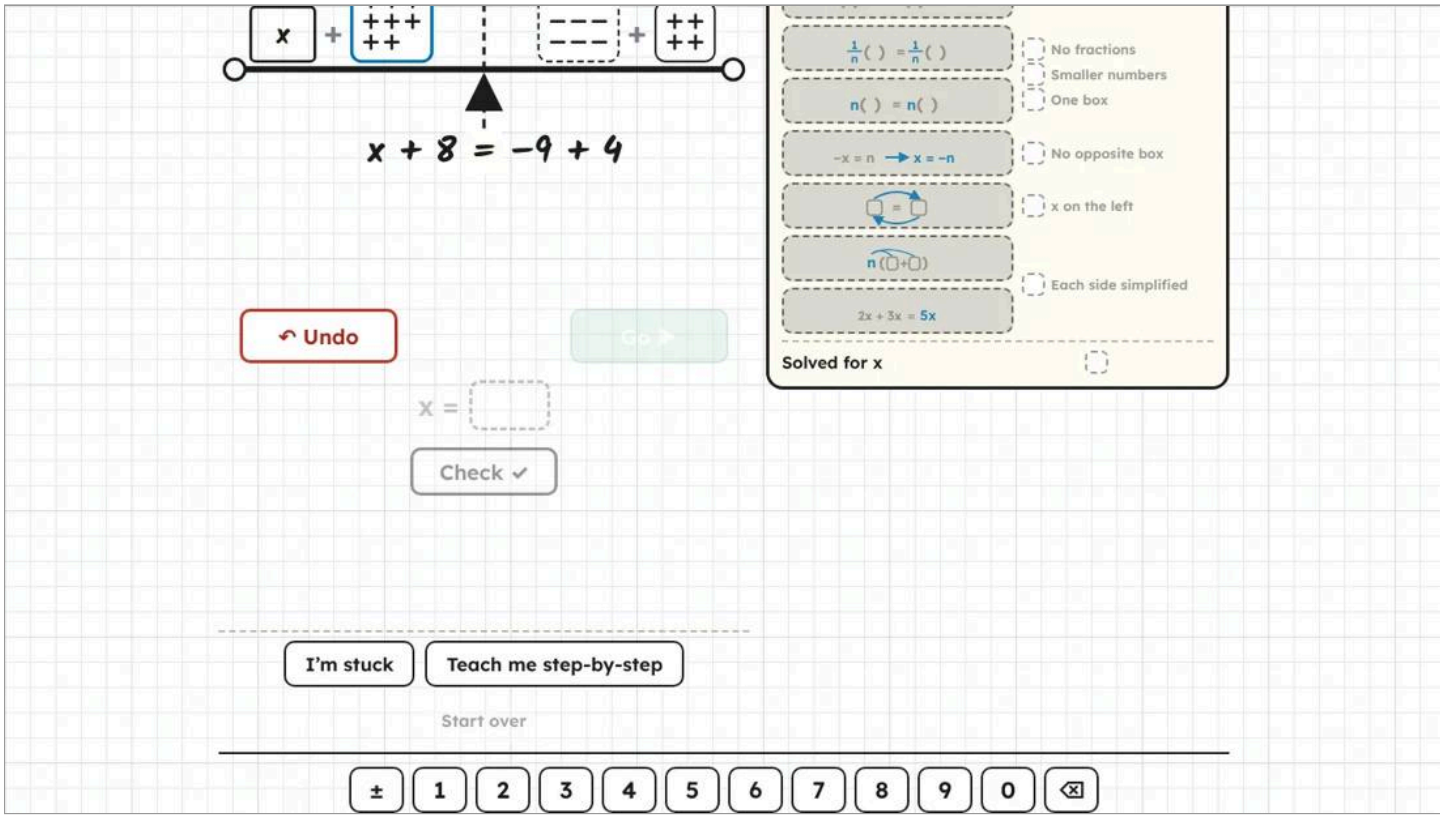
add-18 Add L2 · unhelpful move: Fewer terms red, Undo pulses



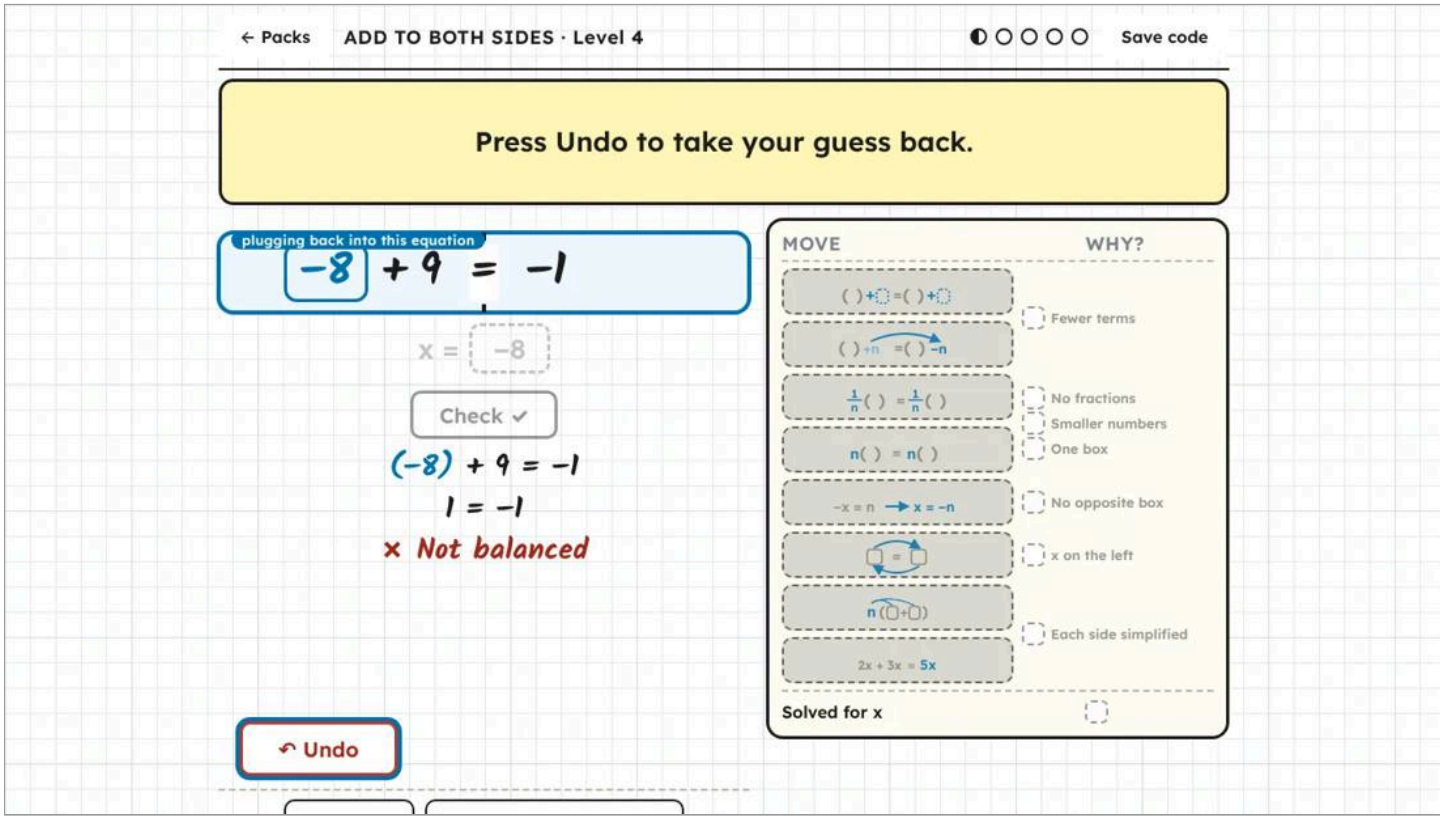
add-20 Add L2 · partly cancelling move: the grey “–” light



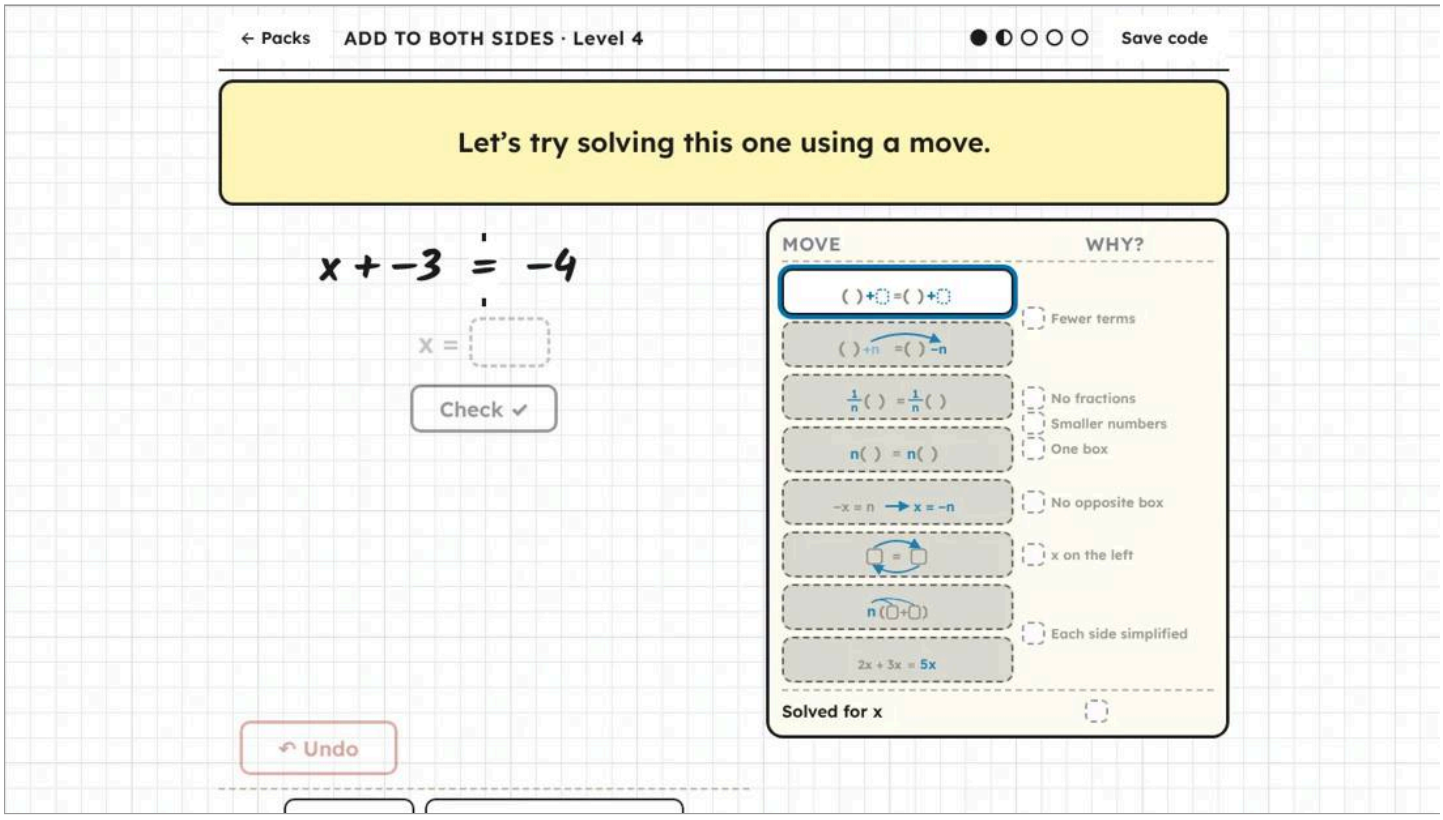
add-26 Add L2 · I’m stuck: the seesaw replaces the work stack



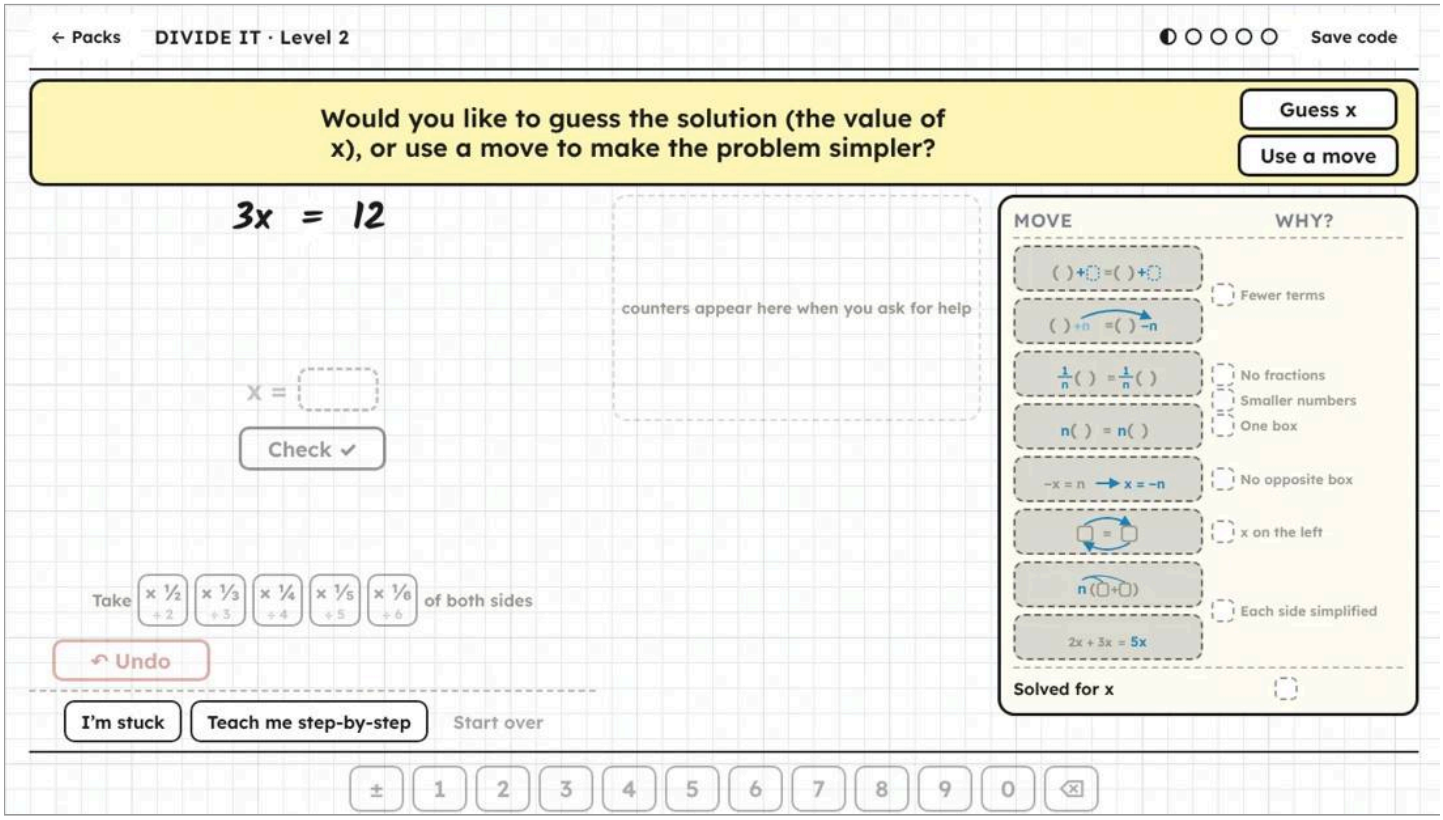
add-42 Add L4 · wrong guess: Not balanced, Undo



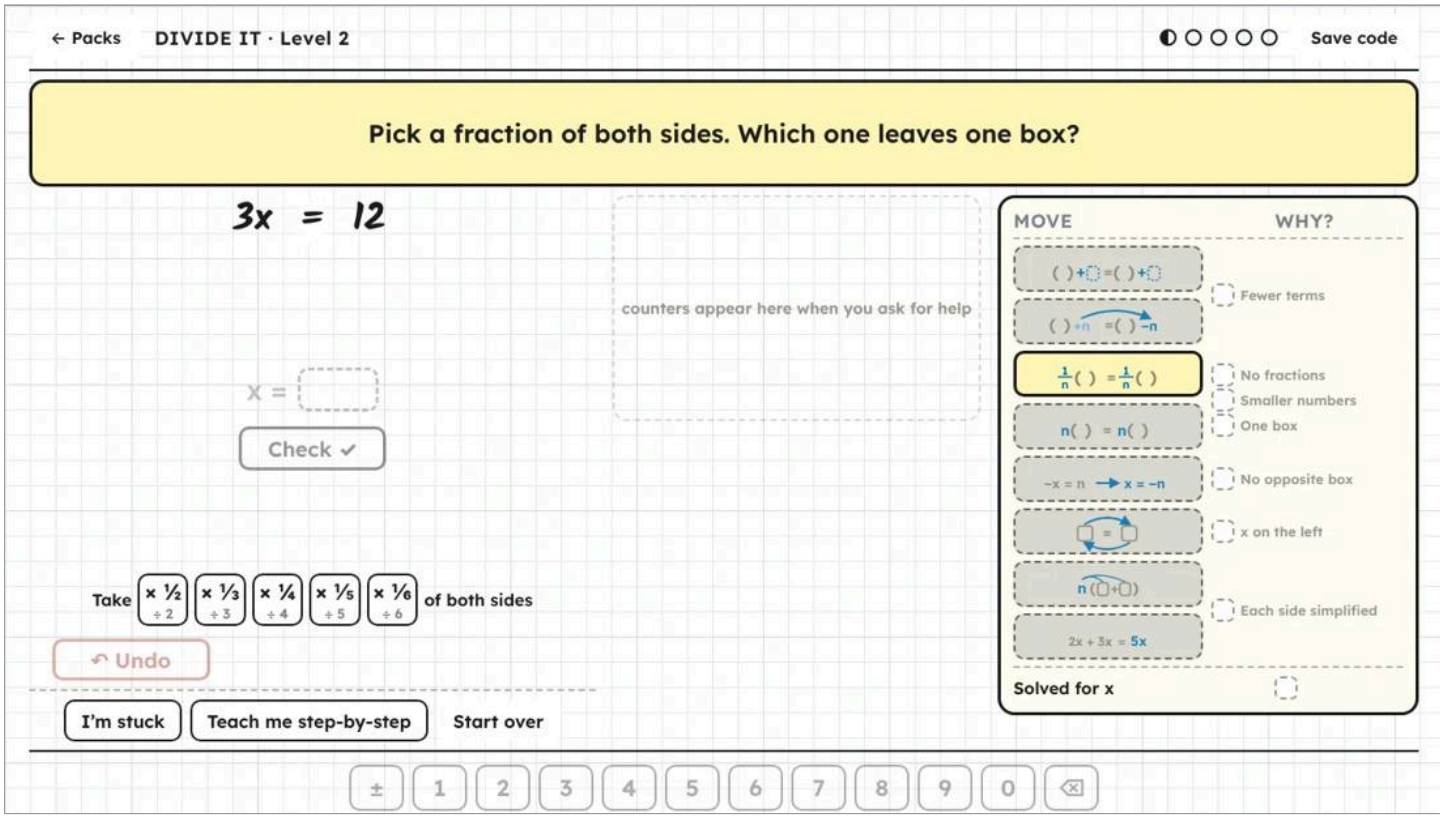
add-45 Add L4 · forced move after a correct guess



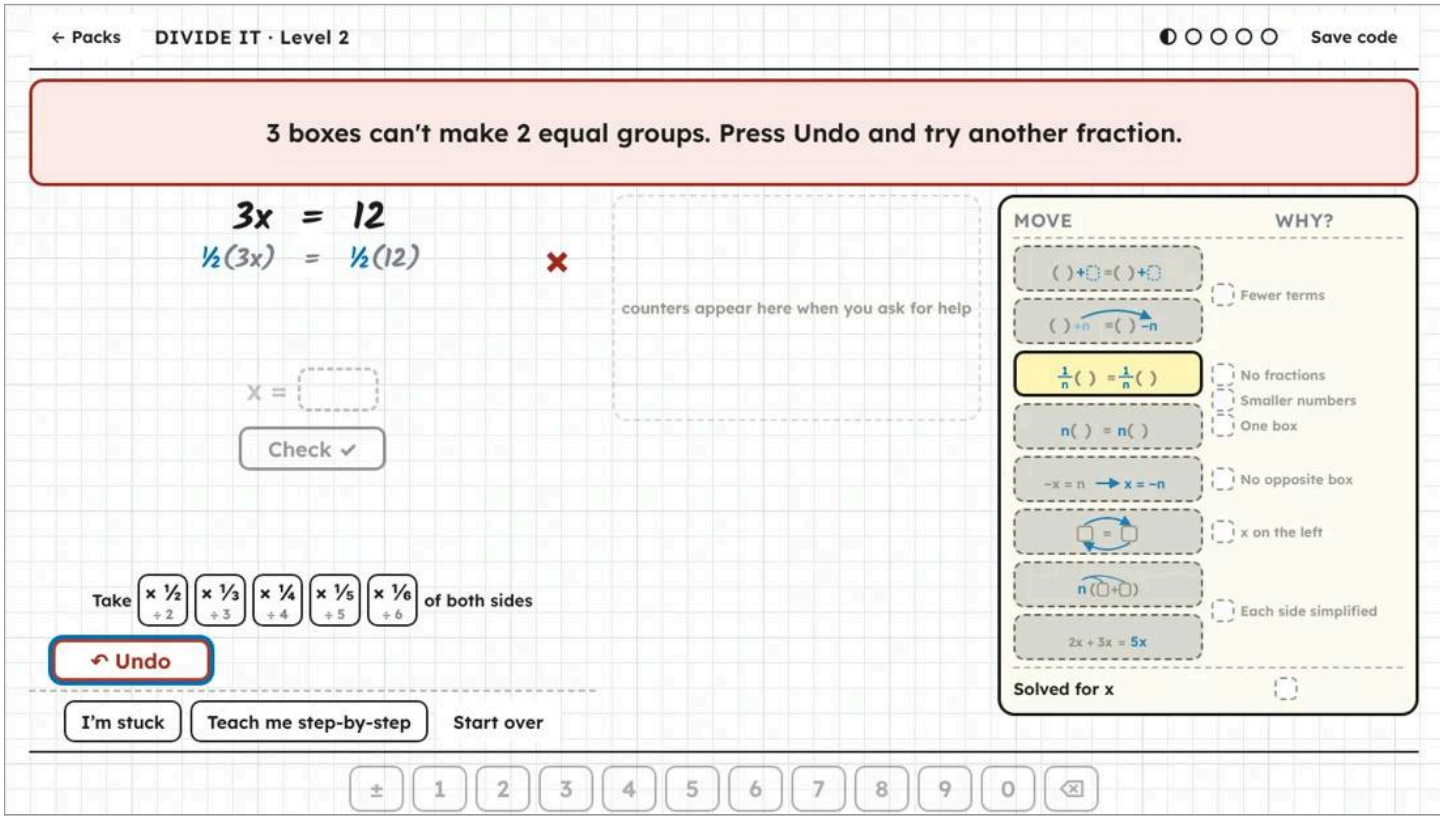
`div-01` Divide L2 · opening (classic layout)



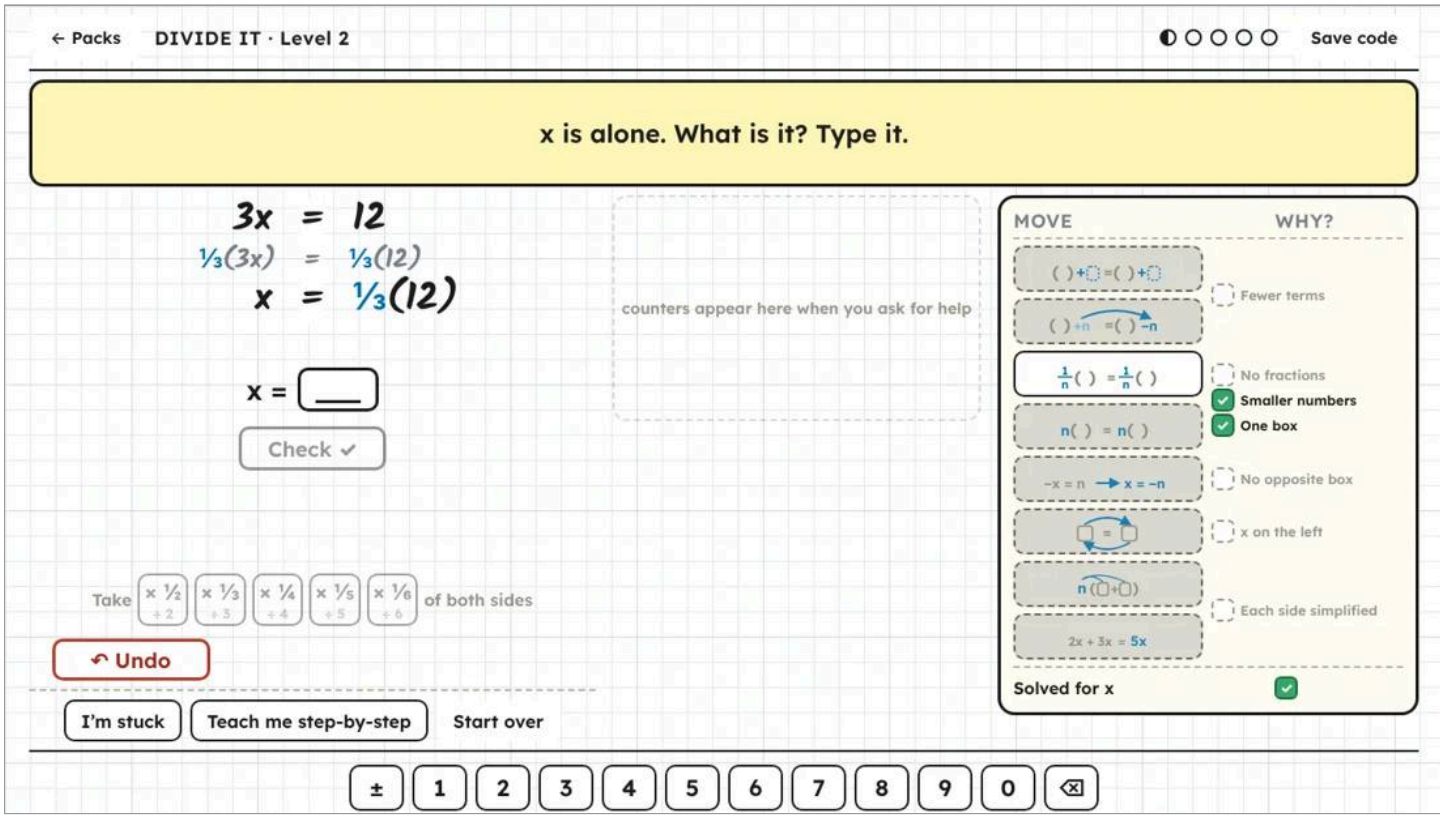
`div-06` Divide L2 · move chosen: the fractions row goes live



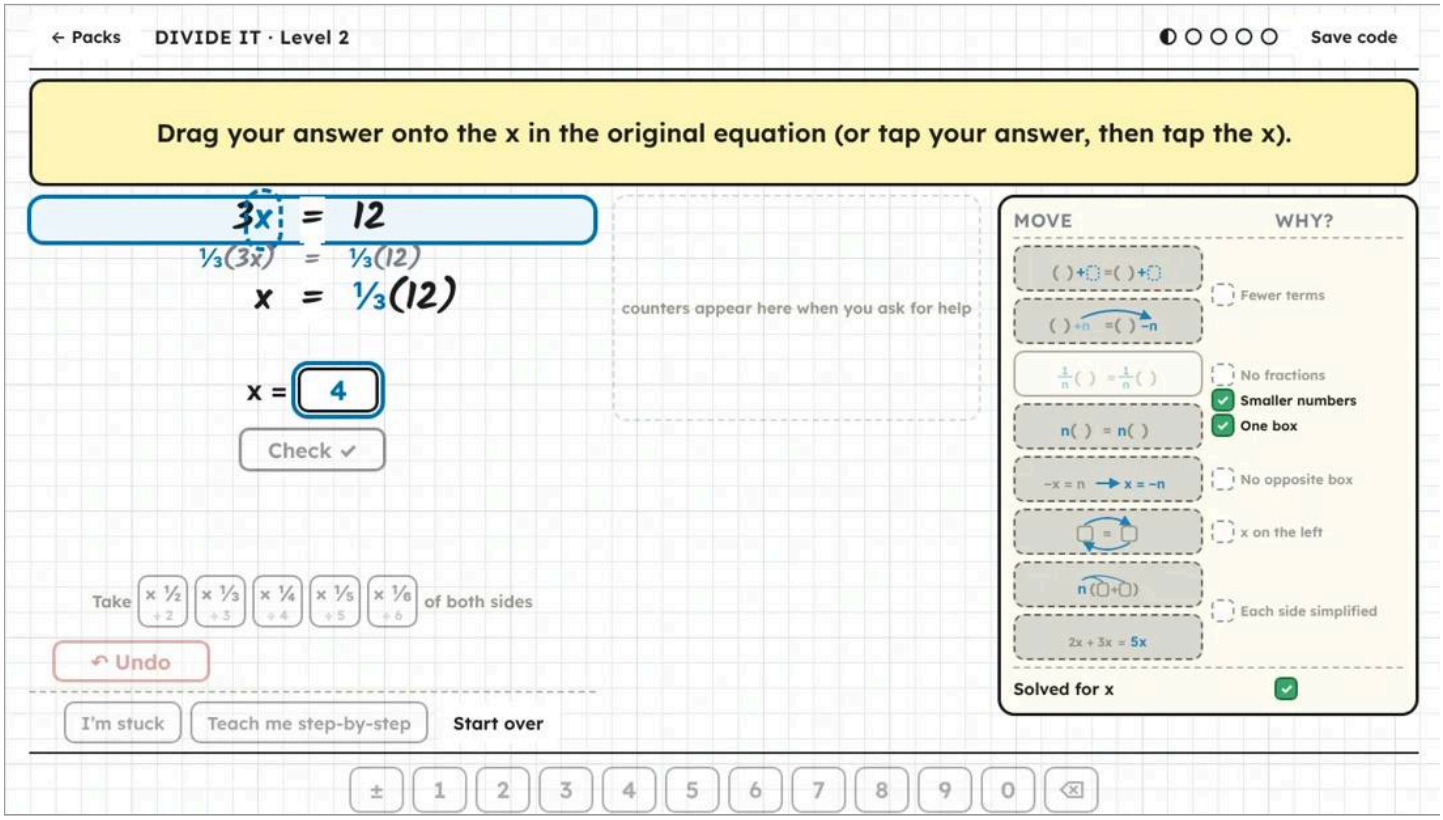
`div-08` Divide L2 · a fraction that does not split



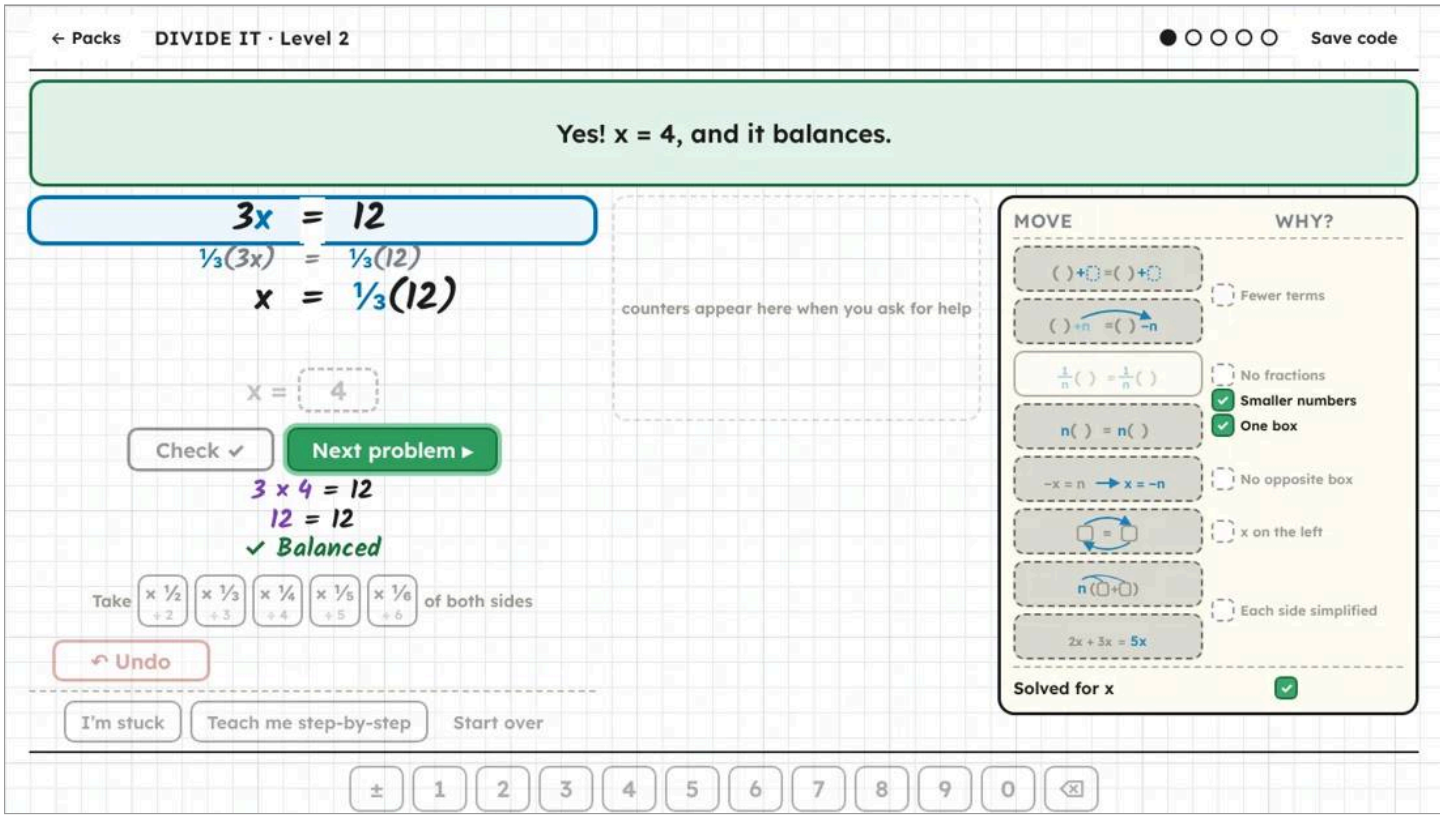
`div-10` Divide L2 · helpful move: x = ⅓(12), lights green



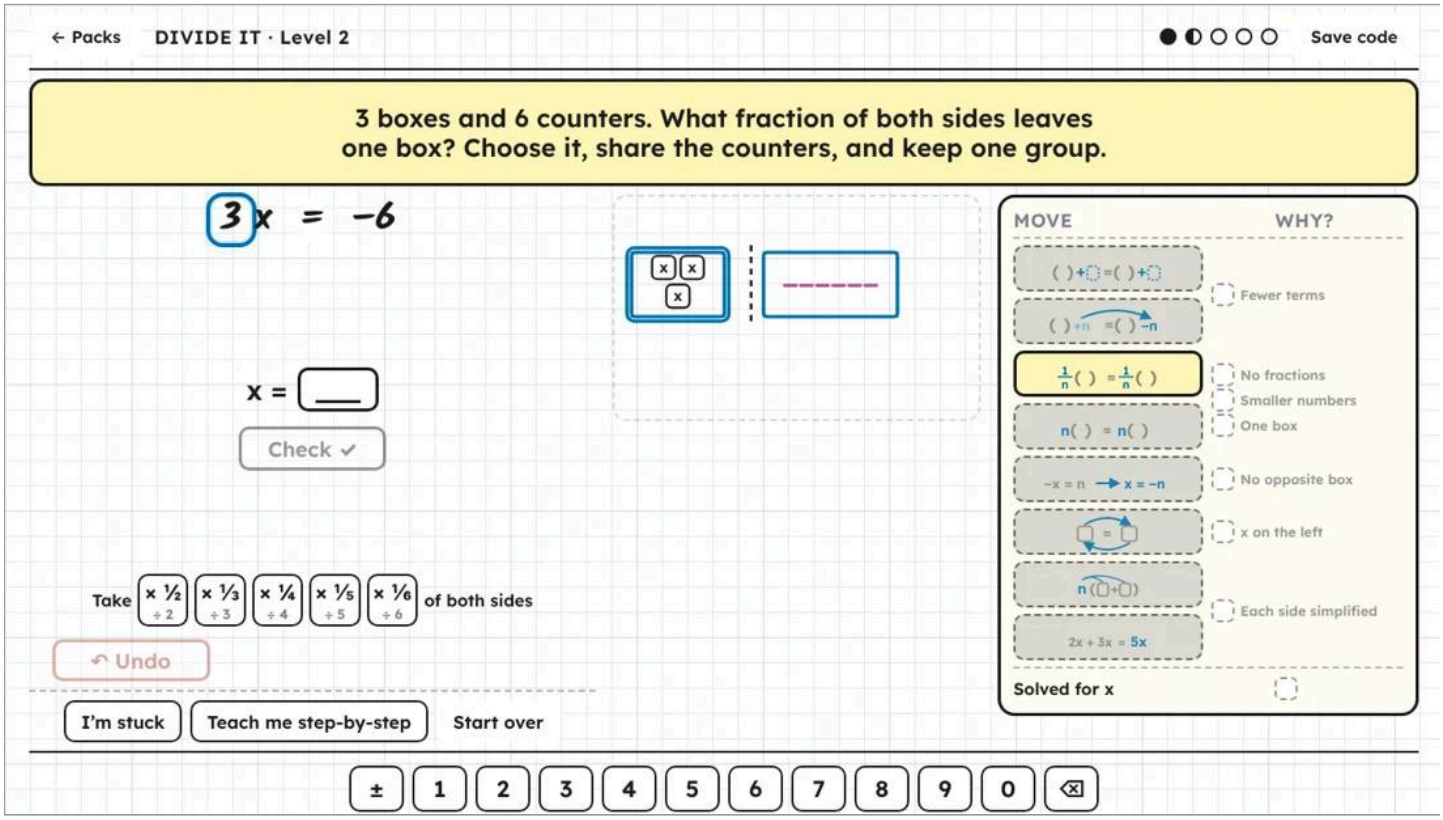
`div-12` Divide L2 · drag the value onto x



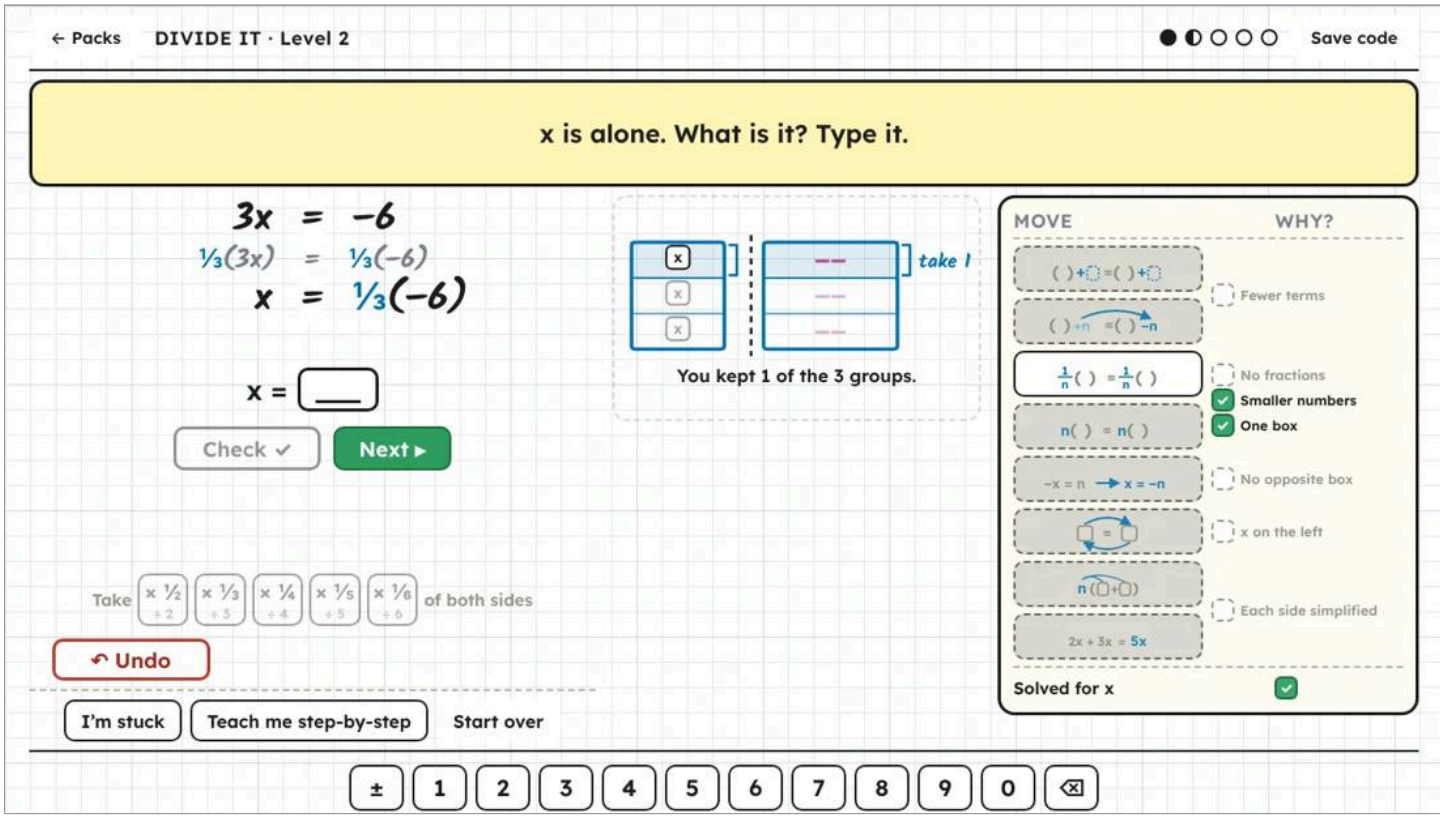
`div-17` Divide L2 · Balanced, end of problem



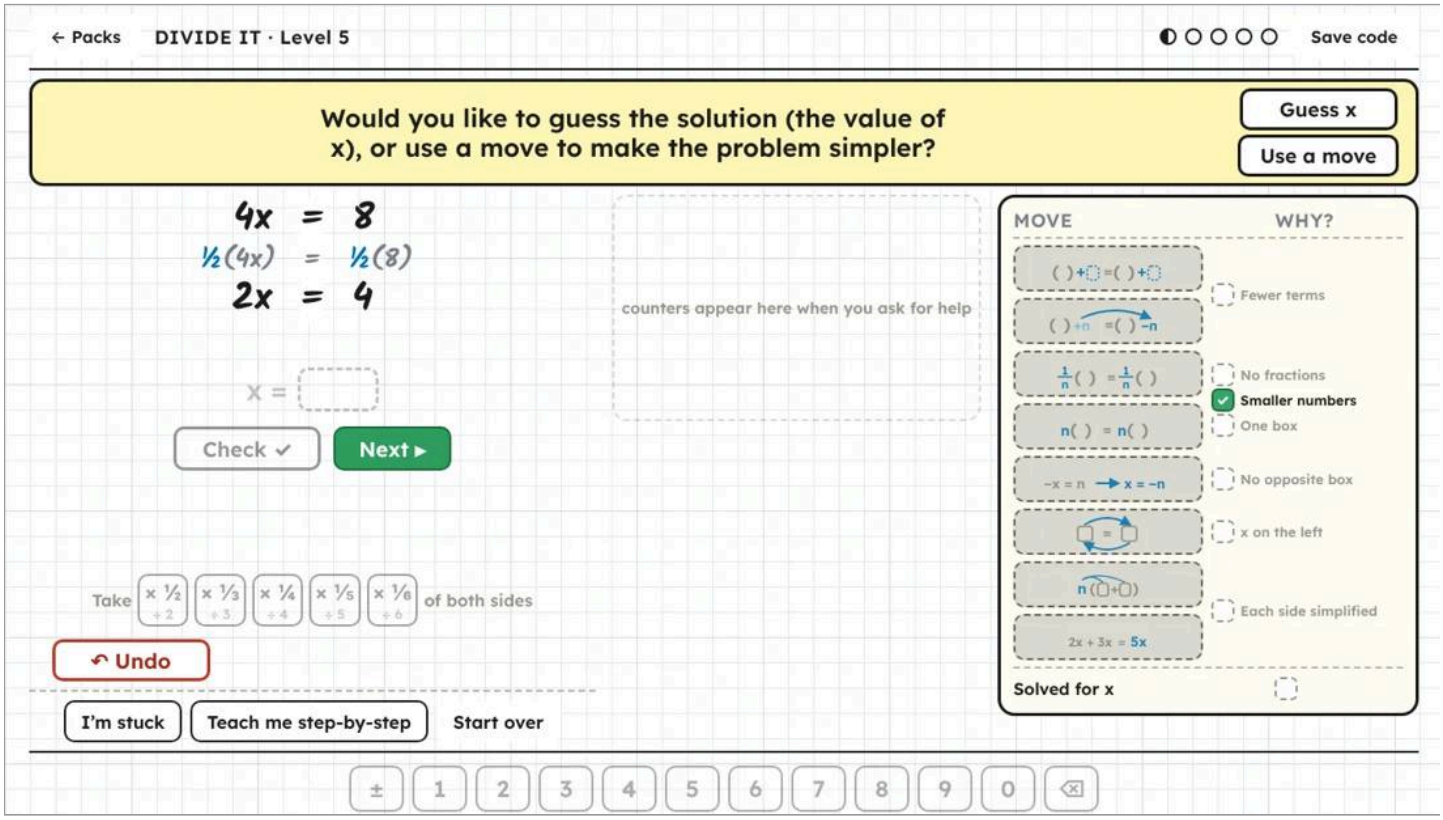
`div-25` Divide L2 · I'm stuck rung 1: counters in the reserved area



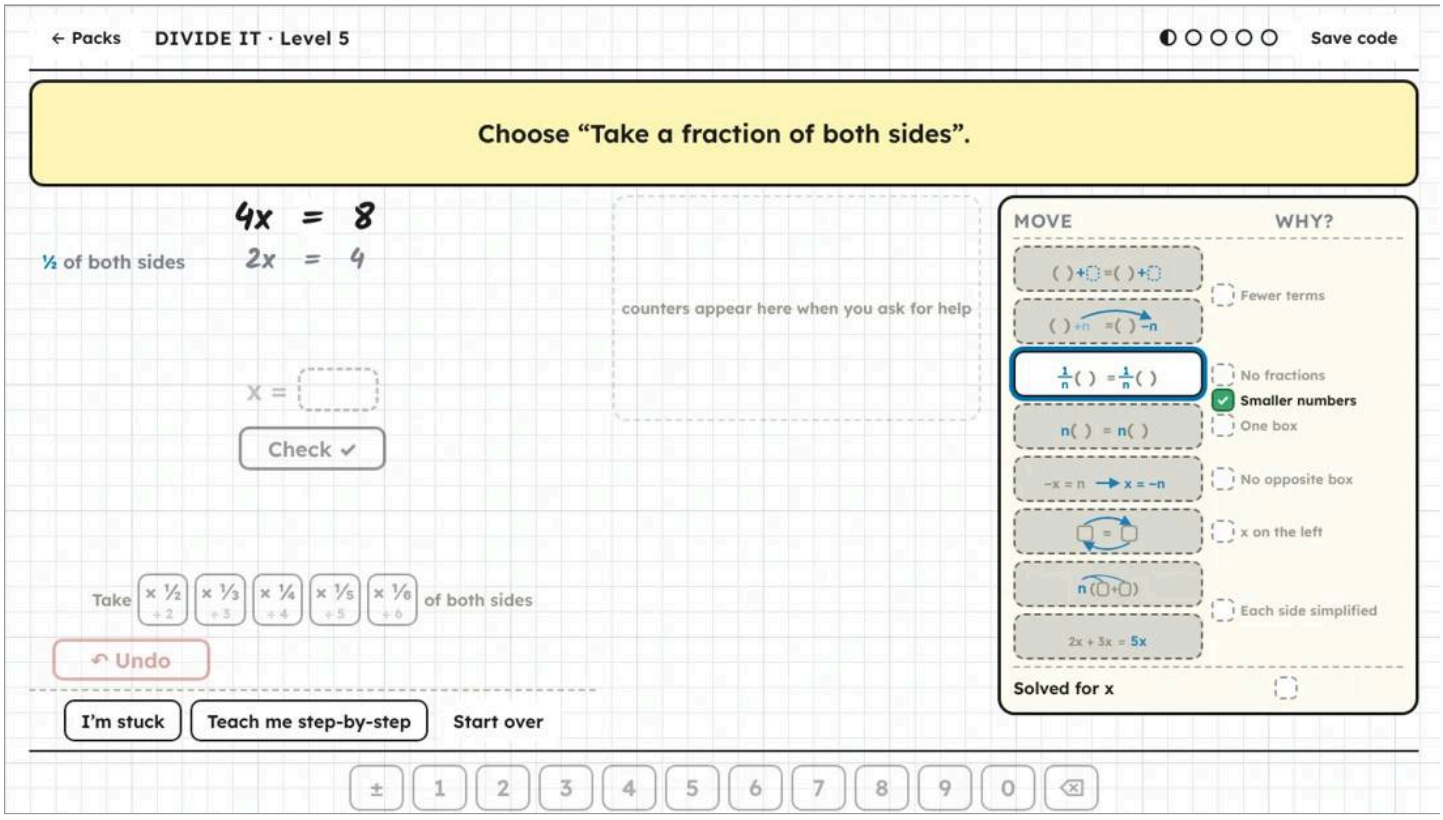
`div-28` Divide L2 · group kept; picture stays until Next



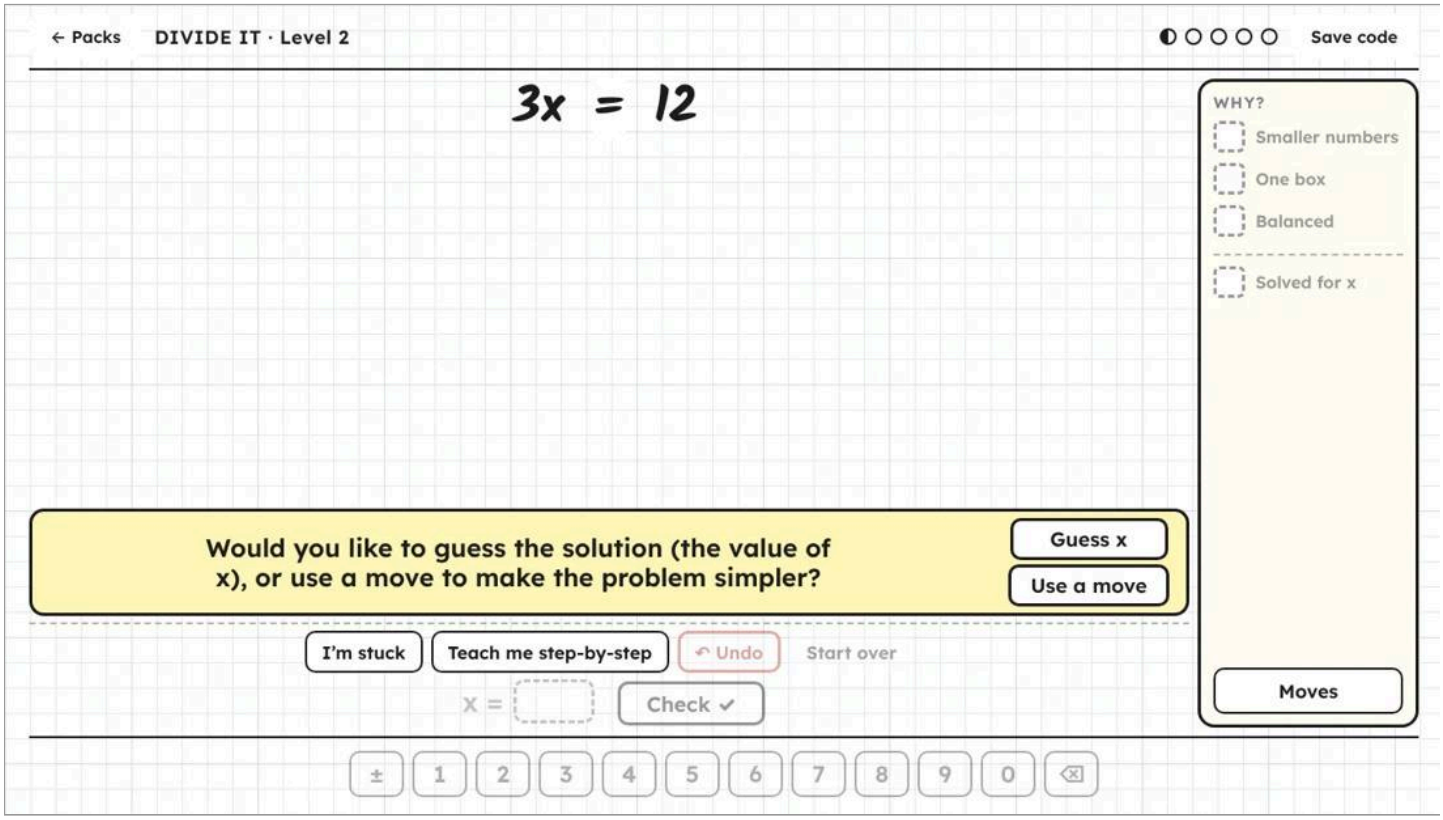
`div-36` Divide L5 · partial move: the choice returns, Next to fold



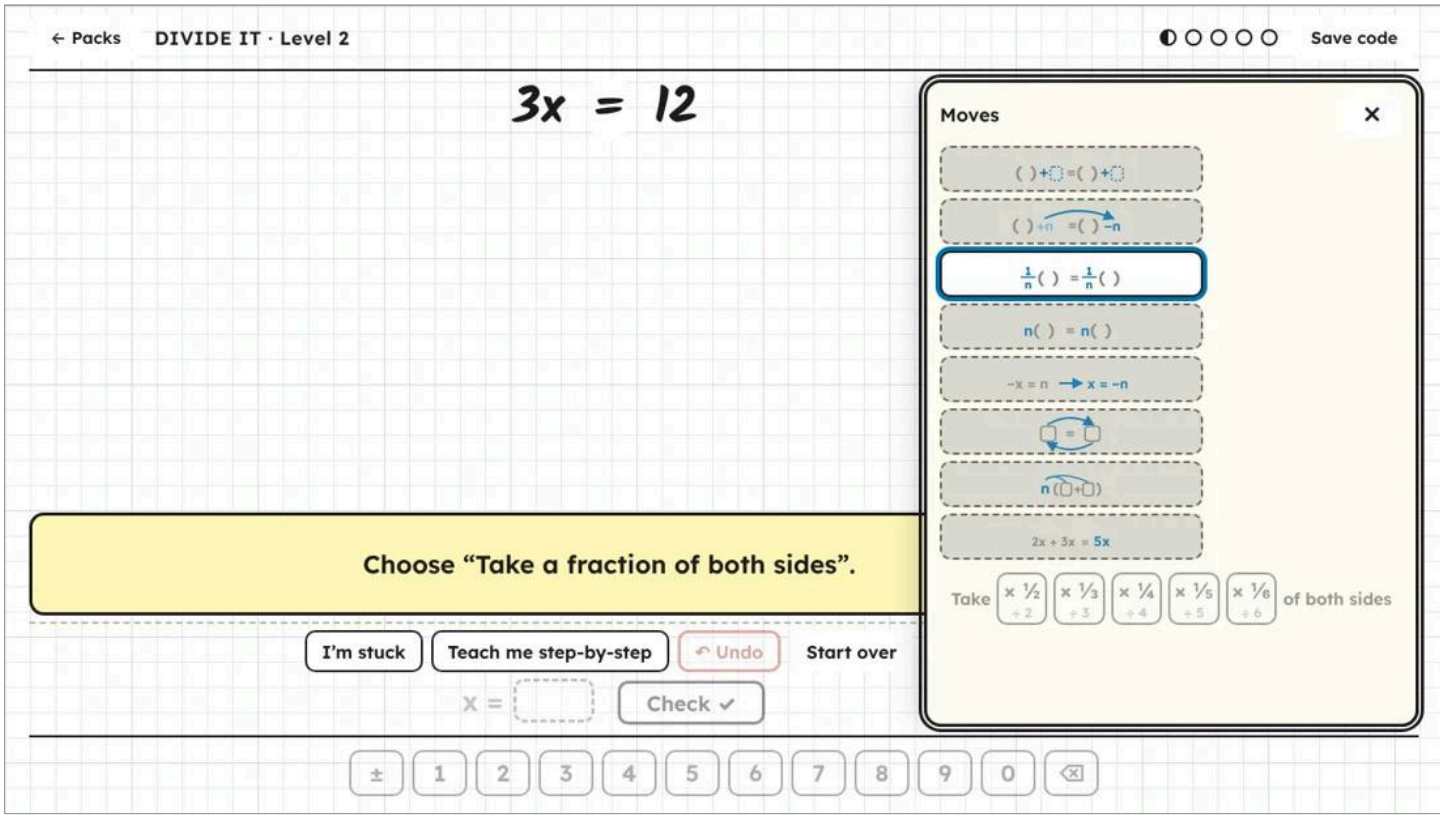
`div-37` Divide L5 · the partial move folded to one row



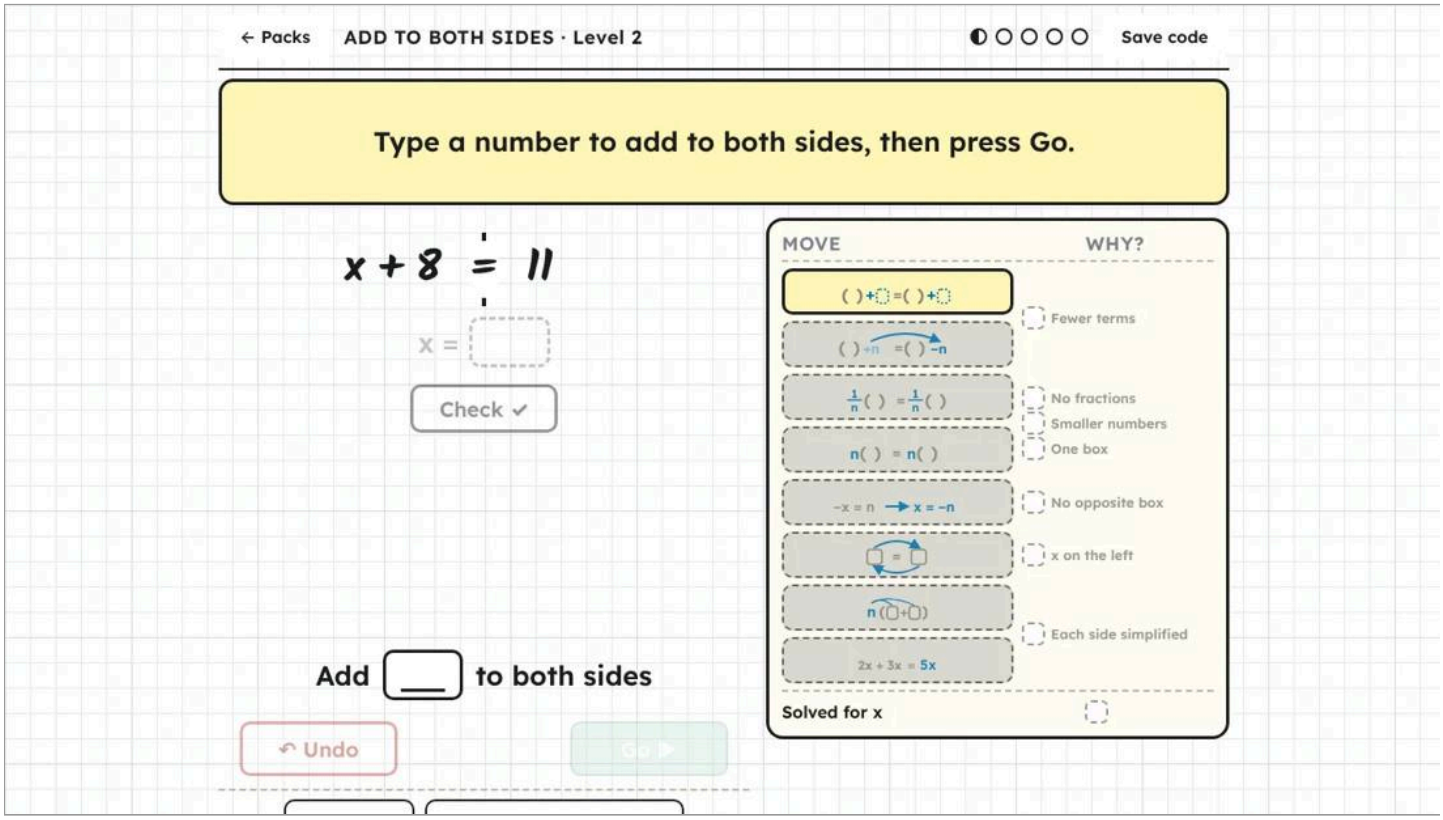
divt-01 Divide tucked pilot · opening



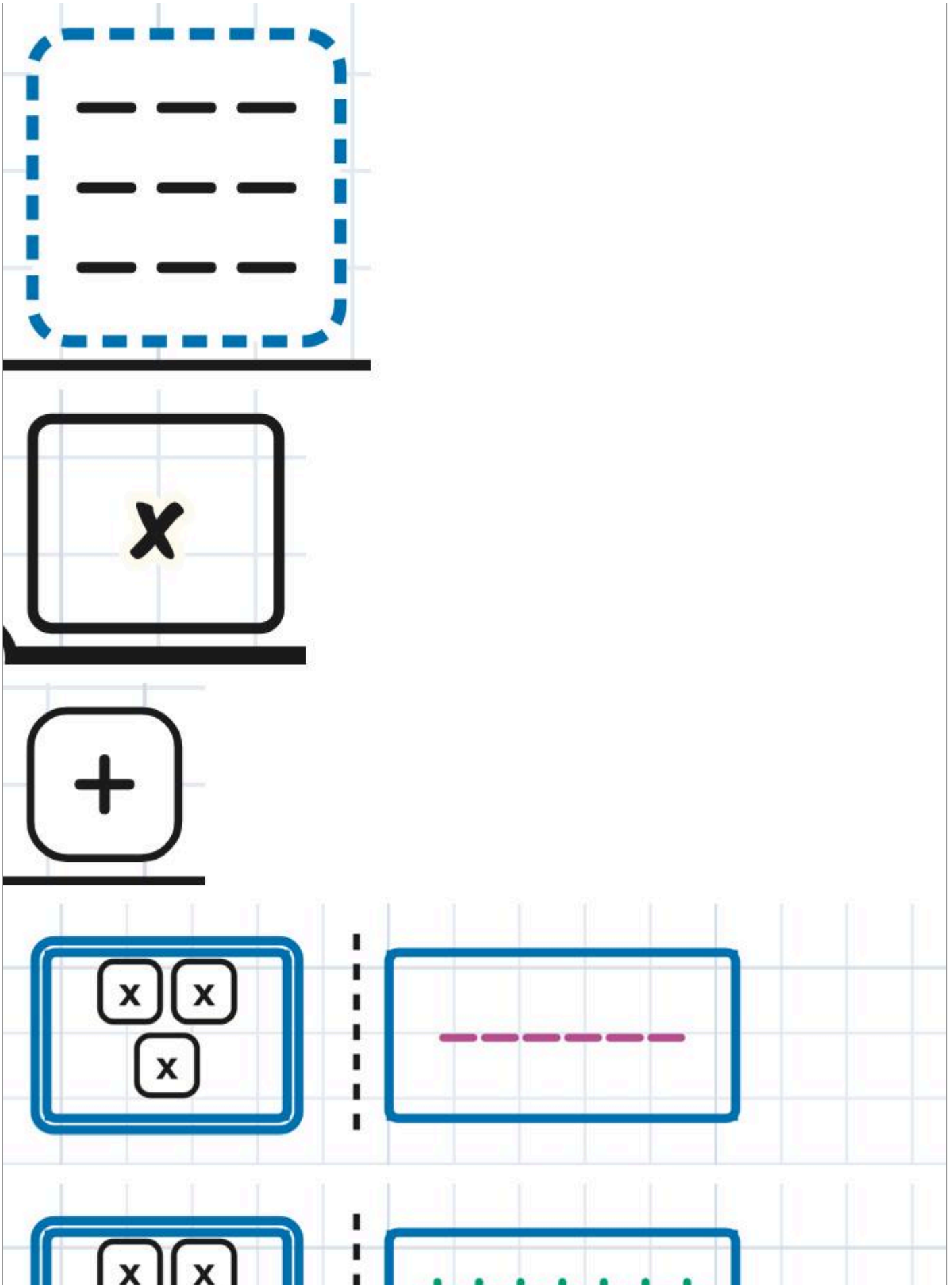
divt-02 Divide tucked pilot · the Moves drawer

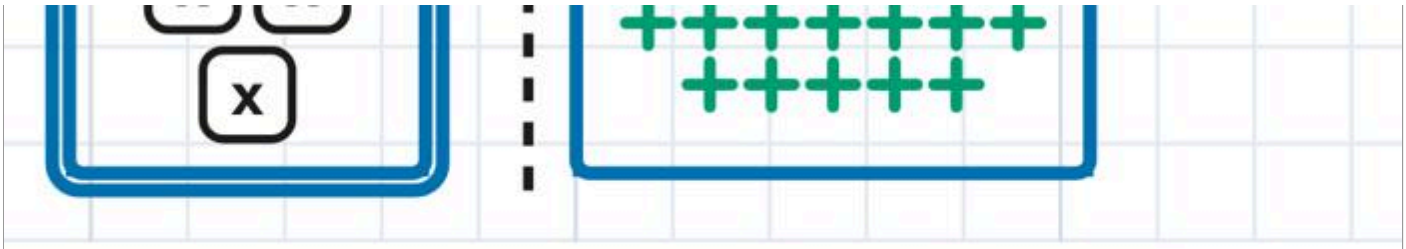


touch-01 Touch · one tap selects the move at once



The pieces at 3× (lights, move icons, counters, x box)





Fewer terms

Fewer terms

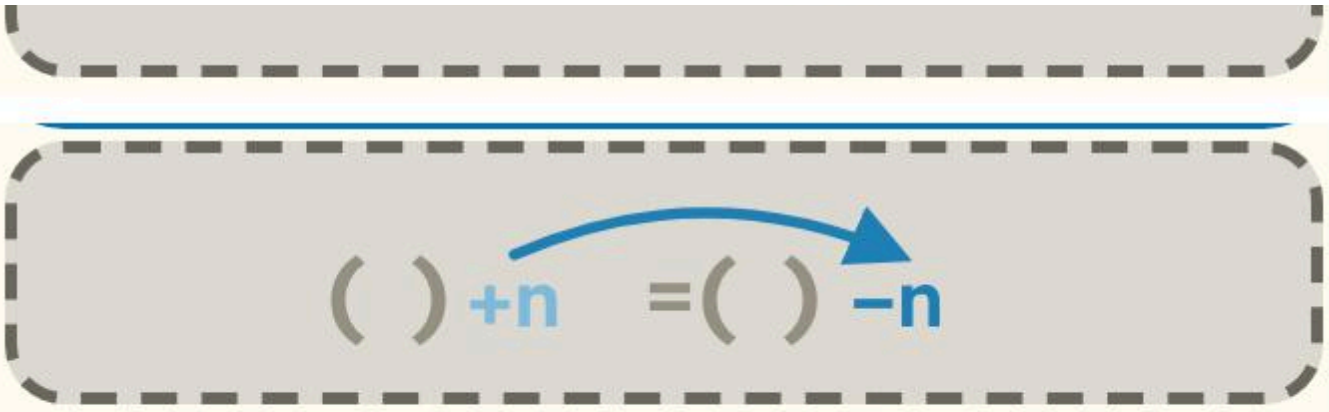
Fewer terms

Fewer terms

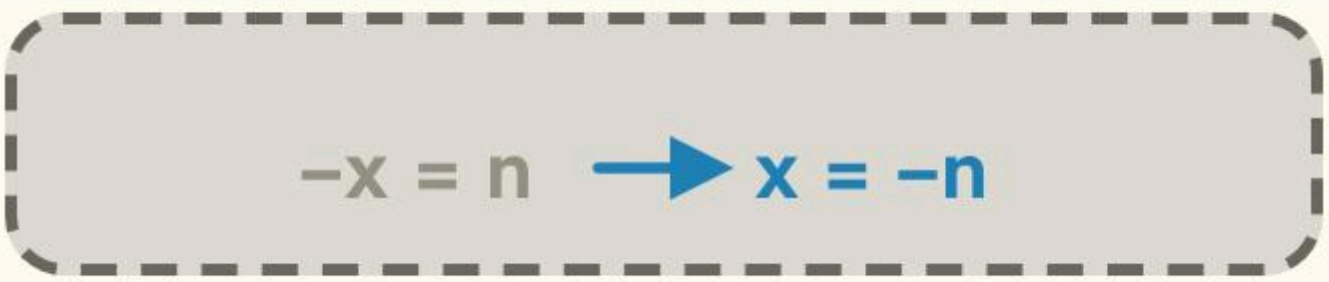
Solved for x

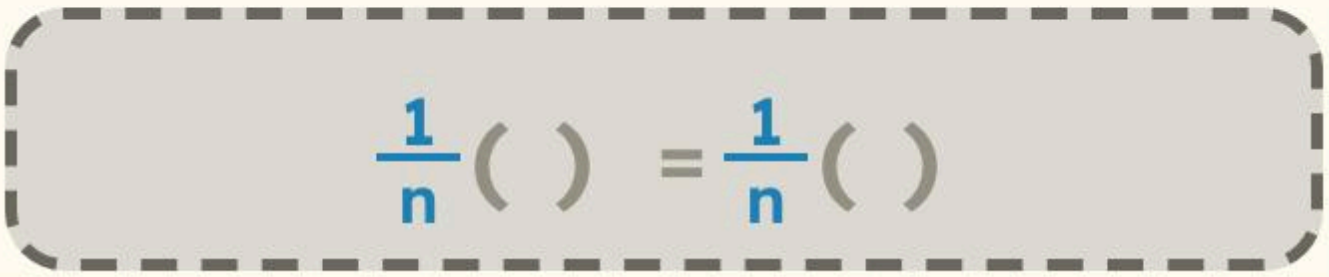
( ) + ☐ = ( ) + ☐

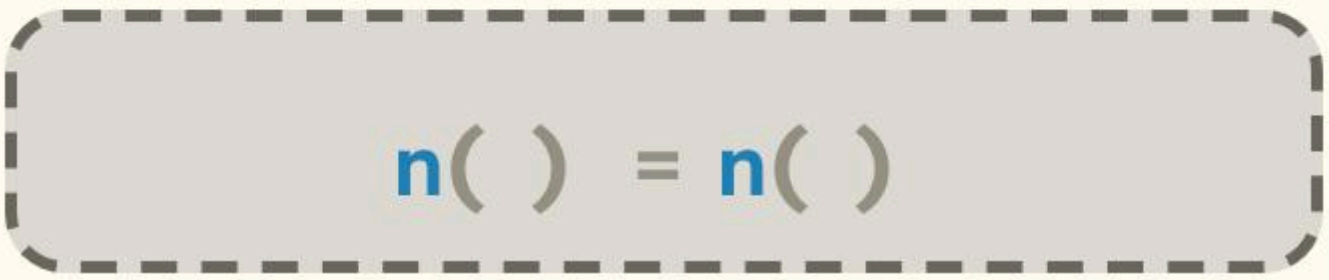
2x + 3x = 5x

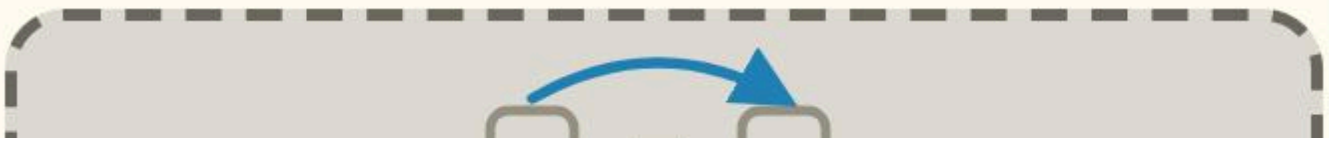

( ) +n =( ) −n

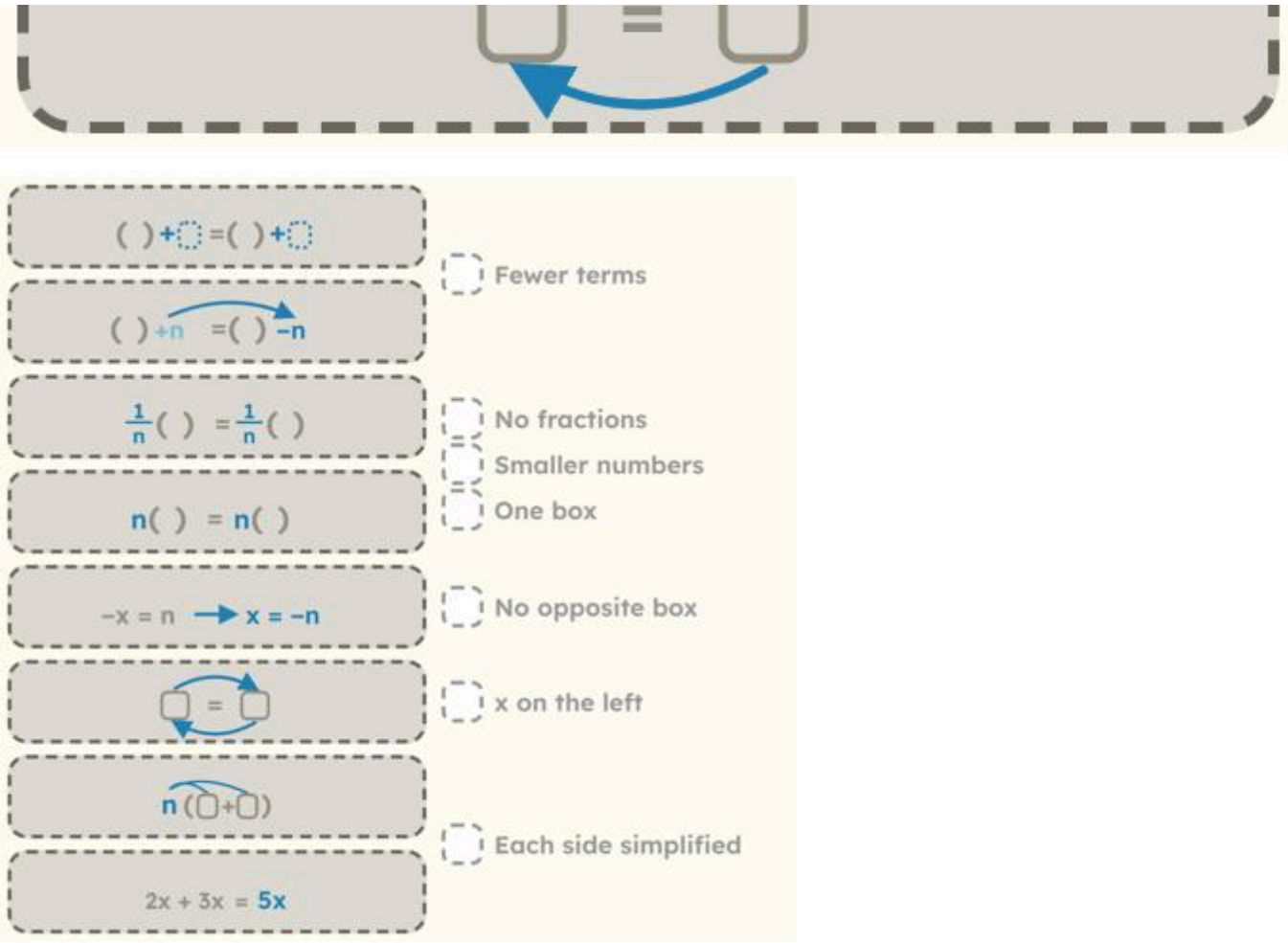

n (□+□)


−x = n → x = −n


1/n ( ) = 1/n ( )


n( ) = n( )





1.2 Layout map at 1366×768

Positions are measured (`measure.json`): x, y, width × height in px from the top-left of the page. "Reserved" means the space is kept when empty.

Add to both sides, Level 2, opening (`add-01`)

The game is a centred 960-px column (x 203 to 1163). The page is **955 px tall** in a 768 window.

Region	x, y	w × h	Reserved when empty?	What fills it
Header (Packs, title, dots, Save code)	203, 8	960 × 54	always there	
Level dots	917, 22	112 × 16		five dots
The line	203, 70	960 × 157	min 120; grows to 157 with the two-line prompt + buttons	prompt, Guess x / Use a move, messages, Undo cue
Work stack (original equation first)	203, 253	504 × 380	**yes, 380 reserved**	equation 70 tall, work rows, x = row, check

Region	x, y	w × h	Reserved when empty?	What fills it
x = [ ] row	384, 329	142 × 56	inside the stack	
Check (with Next beside it when stepping)	385, 397	140 × 46	inside the stack	
Substitution list	455, 451	— × 108	**yes, 108**	check rows
"Add [ ] to both sides"	203, 645	504 × 56	**yes, kept invisible until Add is chosen**	the amount blank
Undo + Go row	203, 713	504 × 56	yes	Undo; Go only after Add is chosen
Help row	203, 781	504 × 108	yes	I'm stuck, Teach me, **Start over wraps to a second line**
Move / Why? panel	723, 239	440 × 495	yes	8 moves × 44 px + gaps, Why? column, Solved for x
Number pad	203, 897	960 × 58	yes	
Scaffold area	none on the bare screen		**not reserved**	I'm stuck **replaces the work stack** with the seesaw (504 × 470) and the page grows to 1,100

Divide it, Level 2, opening, classic (`div-01`)

Full width (1320 px). The page is exactly 768 px tall.

Region	x, y	w × h	Reserved?	What fills it
Header	23, 8	1320 × 54		
The line	23, 70	1320 × 102	min 102	prompt text centred; Guess x / Use a move stacked on the right
Work stack	23, 182	538 × 350	min 300	equation 44 tall, rows, x = row, Check, substitution (84 reserved)
Fractions row "Take × ½ … of both sides"	23, 536	538 × 58	**yes, greyed until the move is chosen**	five fraction buttons
Undo	23, 598	538 × 44	yes	
Help row	23, 650	538 × 50	yes	I'm stuck, Teach me, Start over on one line
Counters / diagram area	577, 180	350 × 215	**yes, an empty dashed box** "counters appear here when you ask for help"	the group diagram

Region	x, y	w × h	Reserved?	What fills it
Move / Why? panel	943, 180	400 × 495	yes	
Number pad	23, 708	1320 × 58	yes	

Divide it, `?layout=tucked` (`divt-01`)

Region	x, y	w × h	Notes
Original equation (pinned)	23, 70	1102 × 58	
Work stack (scrolls)	23, 134	1102 × 338	shrinks to 120 when the tray is open
The line	23, 478	1102 × 102	**moved below the stack**
Helper row (I'm stuck, Teach me, Undo, Start over)	23, 586	1102 × 48	one row
x = row + Check	426, 640	296 × 46	under the helper row, out of the stack
Why? column	1133, 70	210 × 616	lights only, with a Moves tab at its foot
Moves drawer	887, 70	470 × 616	slides over the Why? column, only when opened
Counters tray	23, 470	1102 × 220	slides up only for I'm stuck / Teach me; the diagram inside is 194 tall and **cut off** at the bottom

How much vertical space would be freed

- **The Move panel and the scaffold area sit beside the stack, not above it**, so on both cards removing them frees **width** (Add: 456 px; Divide: 366 px for the diagram box and 416 px for the panel), not height. *Inferred from the x positions above.*
- **What costs height is the column under the stack.** On Add: the reserved "Add [ ] to both sides" row (56 + 12 gap), the Undo/Go row (56 + 12) and the two-line help row (108 + 12) add up to **about 256 px**, more than the 187 px Add overflows by. On Divide: the fractions row (58 + 4), Undo (44 + 8) and the help row (50 + 8) are **about 172 px**. Ideas 5 and 6 (blanks in the equation, Undo in a corner, help beside x =) would remove most of that. *Sums of measured heights and gaps; an estimate.*
- **Add's scaffold** takes no space until asked for, but then it replaces the stack (against loop §6) and adds 145 px to the page.

1.3 How the Move panel and the Why? lights work today

How each Move row is drawn

- **Inline SVG, built in code**, one function per move: `ICONS` and `moveIcon()` in `src/view/lightScreen.js:45-66`. Template: grey `( )` for "whatever is there", the changing part in blue `#0072b2` (`lightScreen.js:32`), a light blue `#6fb6e0` for the "before" of Cross the border. Each sits in a 44-px button, 180-250 px wide (`shots/pieces/move-icon-*.png`).
- **The order** is `MOVES` (`lightScreen.js:17-26`): Add, Cross the border, Take a fraction, Multiply, Flip everything, Trade places, Distribute, Combine.
- **Could the icon be reused as a preview on the equation?** Not directly. The icons are generic pictures (`( ) + ▢ = ( ) + ▢`); the equation rows are **HTML text**, not SVG (`equationRow` and `termsOf` in `src/view/solveScreen.js` and `lightScreen.js`). A preview would be a new "ghost row" drawn with the same HTML row builder, with dashed boxes for the blanks. The **look** (dashed blue box, blue `n`, `1/n`) can be matched in CSS so the icon and the preview read as the same thing. *Inferred from the code.*

What happens when a move is chosen

Add to both sides (bare screen): 1. The line asks **Guess x** or **Use a move**. Until one is tapped the moves are **locked**: greyed, no tooltip (`add-01`, `add-03`). 2. **Use a move**: Add pulses blue (`add-06`). 3. Tap Add: the reserved row **"Add [ ] to both sides"** becomes visible under the stack with a blinking blank; **Go** appears greyed in the Undo row (`add-09`; `lightScreen.js:68`). 4. Type the amount on the pad or keyboard (± toggles the sign). Go turns green and pulses (`add-10`). A number can also be dragged from the equation into the blank. 5. **Go** writes the work row `x + 8 + −8 = 11 + −8` (the amount in blue on both sides, x's pair struck through), then the new row `x = 11 + −8`. The other side is **left unsimplified** so the answer is not shown. The lights are judged (`addBothMoves.js:393` onward, `lightBoard` at `:231`). 6. If x is alone, the `x =` row goes live (`add-11`). There is **no Next and no folding** after a move on this card. Undo stays live for the move. 7. An unhelpful amount: Fewer terms red, the line explains, Undo pulses (`add-18`). Undo keeps Add chosen and clears the blank (`add-19`).

Divide it (on the loop): 1. Same choice; moves locked until **Use a move** (`div-01`, `div-03`). 2. Tap *Take a fraction of both sides*: the reserved **fractions row** under the stack goes live (`div-06`). 3. **Tap a fraction: it is applied at once to both sides.** There is no blank and no Go on this card (`divideMoves.js:109`). 4. A fraction that does not split writes a red row with ×, the line names the side, Undo pulses (`div-08`). Undo returns to the same state with the choice showing (`div-09`). 5. A finishing move writes `⅓(3x) = ⅓(12)` then `x = ⅓(12)`; One box, Smaller numbers and Solved for x turn green; `x =` goes live (`div-10`). 6. A partial move (`4x = 8`, `× ½ → 2x = 4`): the choice returns at once with the same prompt, and a green **Next** appears beside Check (`div-36`). The move **folds** to one row ("½ of both sides 2x = 4") on the student's next action (`div-37`).

Buttons involved: Add uses Go + Undo; Divide uses no Go, Undo, and Next (fold) after a partial move. Both use Check, then Next once per substitution line, then "Next problem".

The Why? lights: every light, its rule, its states, its move

The lights sit in a grid beside the moves they go with (`WHYS`, `lightScreen.js:253-259`); each group is centred on its moves' rows. **Light states that exist in code:** `off` (dashed outline), `green` ✓, `red` ×, `neutral` (grey –). **There is no partly state.**

Light	Beside (pairing, `lightScreen.js:253-259`)	Add to both sides	Divide it
Fewer terms	Add, Cross the border	**Rule:** `judge()` `addBothMoves.js:84-88`. Terms on x's side are counted as `xTerms` = x, plus 1 if the numbers there add to anything but 0 (`:80`). Green if fewer than before; red if more, or if the number next to x got bigger in size; grey (neutral) otherwise. States: off, green, red, neutral.	forced `off` (`divideScreen.js:92`)
No fractions	Fraction, Multiply	no rule: **copies Fewer terms' green** (`lightScreen.js:287`)	forced `off`
Smaller numbers	Fraction, Multiply	no rule: copies Fewer terms' green	**green after any successful fraction**, a constant (`divideMoves.js:227`); never red; a failed fraction leaves it as it was
One box	Fraction, Multiply	no rule: copies Fewer terms' green	green when the coefficient of x is 1 (`divideMoves.js:227`)
No opposite box	Flip everything	no rule: copies Fewer terms' green	forced `off`
x on the left	Trade places	no rule: copies Fewer terms' green (Trade places is live on L5, but this light is never judged)	forced `off`
Each side simplified	Distribute, Combine	no rule: copies Fewer terms' green	forced `off`
Solved for x	its own row, last	green when x is the only term on its side (`judge`, `:87`)	green when x is alone (`divideScreen.js panelBlock`)
Balanced (tucked pilot only)	Why? column	—	off; green or red after a check (`divideScreen.js:115-123`). Not in any spec.

When they are judged: only after a committed move (Add: after Go; Divide: after the fraction is taken). A wrong guess does not change them. Undo restores the lights saved before the move.

Which moves are visible, greyed, hidden or locked

All eight moves are **always drawn** on both cards, at every level (nothing is hidden). What controls it is `movePanel()` (`lightScreen.js:263`) with the card's `active` move and a `locked` flag from each card's `panelBlock`.

	Add to both sides	**Divide it**
Active	Add, every level including L1	Take a fraction, every level including L1
Also live	Trade places from L5 when x is on the right; the first time it is pointed out and Add waits for it (`addBothMoves.js` `TRADE_LEVEL`, `tradePending`)	—
Greyed (tappable, never acts)	the other six; a tap says "That move unlocks on a later card."	the rest; an earlier card's move (Add, Cross, Trade) says "You don't need that move here.", a later one "That move unlocks on a later card."
Locked (active move drawn greyed, no tooltip)	before the choice and in guess mode	before the choice and in guess mode
Why? boxes of greyed moves	dashed outlines (but they turn green with Fewer terms, see above)	dashed outlines

Hover and touch

- **Hover** on a live move shows its name in a dark tooltip to the left of the icon (CSS `::after`, `src/style.css:2943`). Hover on a greyed or locked move shows nothing. **Nothing else depends on hover**; every action is a click or tap.
- **Press**: no pressed style; it looks like hover (`add-08`, `div-05`).
- **Touch** (emulated): one tap selects the move at once; no name is shown (`touch-01`, `touch-02`). The tooltip CSS is not limited to devices that can hover, so on some touch browsers the name may stick after a tap. *Inferred from the CSS.*
- Keyboard: digits, ± and Backspace type into whichever field is live; **Enter** presses the first available of Check, Next, Go (`src/main.js:530-561`).

The x = answer row

- **Add:** `answerBlock()` `lightScreen.js:172`. **Inside the work stack**, under the newest equation, with **Check directly under it** and **Next beside Check** while the substitution steps (`add-13`).
- **Divide:** `answerRow()` `solveScreen.js:113`, same place and order.
- **Live** (white box, blinking cursor) in guess mode, once x is alone, and on Divide while counters are up (`answerOpen`). **Greyed** before the choice, on the move path until x is alone, while a substitution is stepped, and when done. While the value is being placed on x (Divide), the box becomes a chip to drag onto the x.

- **Tucked pilot:** the `x =` row moves **out of the stack** to the bottom of the work column, under the helper row (`divt-01`).

Where I'm stuck, Teach me, Undo and Start over are

- **Add:** Undo under the stack (150 × 52); help row at the bottom of the left column, 504 × 108, **below the fold** at 1366×768, with Start over wrapping to its own line.
- **Divide (classic):** Undo 150 × 40 above the help row (538 × 50), all three on one line; everything visible.
- **Tucked:** one helper row of four (I'm stuck, Teach me, Undo, Start over), 1102 × 48, right above `x =`.
- I'm stuck and Teach me are live at the start of a problem (an option at any time); Start over is greyed on the untouched original on Divide, and on Add until three red moves.

I'm stuck and Teach me (for reference)

- **Add:** click 1 is a reminder (the number next to x outlined, the blank opened; `add-25`); click 2 brings in the seesaw and **replaces the work stack** (`add-26`; at L2-L3 it starts with building the equation); click 3 the two-action move; click 4 the amount pre-filled. Teach me: the seesaw with the amount pre-filled, add to each side, Go, then Party/Battle (`add-31` to `add-34`).
- **Divide:** rung 1 shows the boxes and the pile in the reserved diagram area with the fractions live, and the student deals and keeps (`div-25` to `div-28`); rung 2 highlights the needed fraction (`div-33`). At x alone: the pile to deal, then the groups made with one bracketed (`div-29`, `div-30`). The stack never leaves the screen.

1.4 The look

Fonts (`index.html:16`, computed styles in `measure.json`) - **Lexend** for everything but equations. Loaded weights: **400 and 600 only.** The line asks for 800 at 24 px, Why? labels 600 at 13-14 px, help buttons 600 at 17 px. - **Kalam 700** (handwritten) for equations: 36 px on Add's stack and Divide's stack.

Colours

Thing	Value	Where set
Page	white `#ffffff` with a 24-px grid `#e6ecf3` / every 5th `#d5dee9`	`style.css:10-14, 31-45`
Ink	`#1d1d1d`	`style.css:3`
The line	`#fff6b8` fill, ink border, 14-px corners; red-tinted when a message is "bad", green when done	`style.css:2768`

Thing	Value	Where set
Move / Why? panel	`#fffdf2`, ink border	
Operation blue (amount added, fraction, icons' changing part)	`#0072b2`	`lightScreen.js:32`; `style.css:2846` (`.lb-num.is-added`), `:2970`
The same blue also marks **"click here"** (pulsing rings), the drop target, hint outlines and the plug-back-in row	`#0072b2`	`style.css:2786, 2794, 2834, 2881, 2966`
Step highlight in the check (accent)	purple `#7b3fb0`	`style.css:3044`
Positive counters	**Add seesaw:** ink + marks in a white rounded box. **Divide diagram:** green `#009e73`	`style.css:2609`; `--inside`, `style.css:7`
Negative counters	**Add seesaw:** ink – marks in a **dashed** box. **Divide diagram:** magenta `#b8508f`	`style.css:2610`; `--opposite`, `style.css:4`
x box	white fill, ink stroke 2.5 px, rounded 6; a bold "x" inside	`style.css:3062-3063`
Opposite box	**not drawn by any Solve it card.** Build it's Boxes & Circles draws it in magenta `#b8508f`	`style.css:1342`
Light green ✓	fill `#3aa76d`, border `#1b6e3a`	`style.css:2681`
Light red ×	fill `#d9534f`, border `#a3281b`	`style.css:2682`
Light neutral –	fill `#ececec`, ink border	`style.css:2683`
Light off	white, dashed `#9a9a9a` border, grey label	`style.css:2684, 2679`
Move button, greyed	`#dcdad2` fill, dashed `#6b685f` border, icon at reduced opacity	computed

Inconsistency worth knowing: the two cards draw counters differently (Add: black marks, sign shown by a dashed box; Divide: green and magenta marks).

Pieces at 3× are in `shots/pieces/`: `add-seesaw-x-box`, `add-tray-plus-piece`, `add-seesaw-negative-group-hinted` (the dashed box, shown with its blue hint outline), `divide-counters-positive`, `divide-counters-negative` (boxes and a pile), the four light states, Solved for x green, all eight move icons, and the locked panel. No opposite box exists in Solve it to capture.